warning_system

Design Description

nites

warning_system: Design Description

by nites

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Chapter 1. Model Version

Version: 1.942

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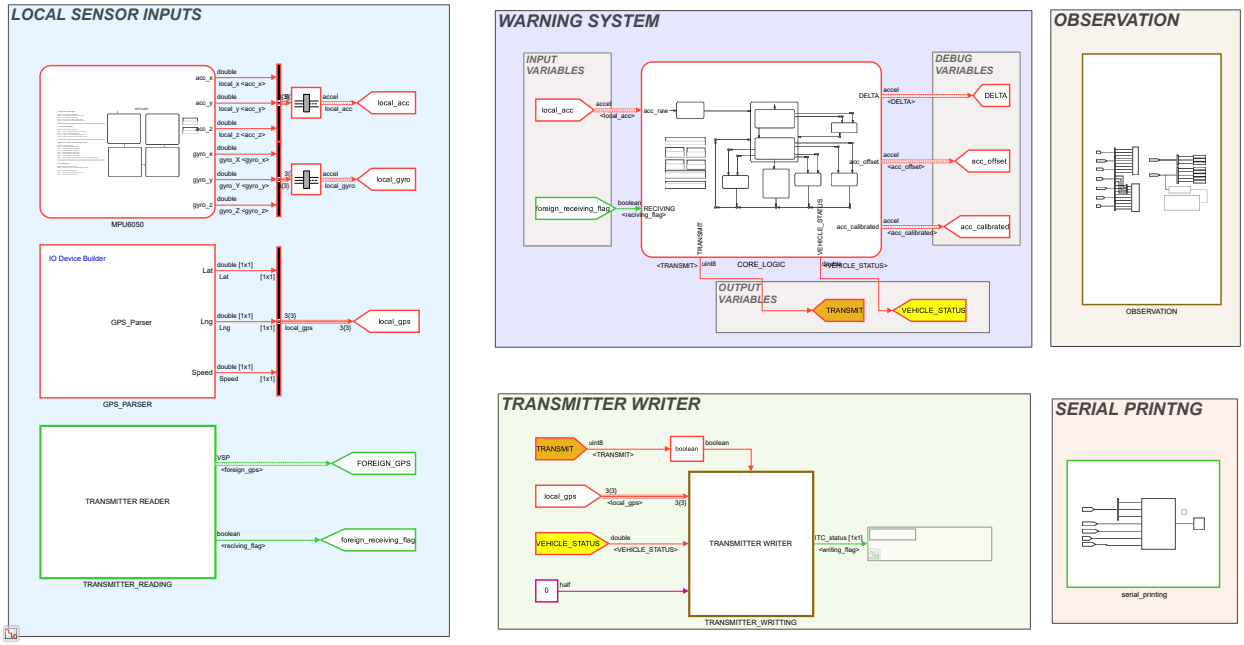
Chapter 2. Root System

Figure 2.1. warning_system



Blocks

Parameters

"Bus Creator" (BusCreator)

Table 2.1. "Bus Creator" Parameters

Parameter	Value
Number of inputs	3
Display option	bar
Data type	Inherit: auto
Output as nonvirtual bus	off
Use names from inputs instead of from bus object	on

"Bus Creator1" (BusCreator)

Table 2.2. "Bus Creator1" Parameters

Parameter	Value
Number of inputs	3
Display option	bar
Data type	Inherit: auto
Output as nonvirtual bus	off
Use names from inputs instead of from bus object	on

"Bus Creator2" (BusCreator)

Table 2.3. "Bus Creator2" Parameters

Parameter	Value
Number of inputs	3
Display option	bar
Data type	Inherit: auto
Output as nonvirtual bus	off
Use names from inputs instead of from bus object	on

"Cast To Boolean" (DataTypeConversion)

Table 2.4. "Cast To Boolean" Parameters

Parameter	Value
Output minimum	[]
Output maximum	[]
Output data type	boolean
Lock output data type setting against changes by the fixed-point tools	off
Input and output to have equal	Real World Value (RWV)
Integer rounding mode	Floor
Saturate on integer overflow	off
Sample time (-1 for inherited)	-1

"Constant1" (Constant)

Table 2.5. "Constant1" Parameters

Parameter	Value
Constant value	0
Interpret vector parameters as 1-D	on
Output minimum	[]
Output maximum	[]
Output data type	half
Lock output data type setting against changes by the fixed-point tools	off
Sample time	inf
Frame period	inf

"From" (From)

Table 2.6. "From" Parameters

Parameter	Value
Goto tag	local_acc
Icon display	Tag

"From2" (From)

Table 2.7. "From2" Parameters

Parameter	Value
Goto tag	local_gps
Icon display	Tag

"From6" (From)

Table 2.8. "From6" Parameters

Parameter	Value
Goto tag	foreign_receiving_flag
Icon display	Tag

"From7" (From)

Table 2.9. "From7" Parameters

Parameter	Value
Goto tag	TRANSMIT
Icon display	Tag

"From8" (From)

Table 2.10. "From8" Parameters

Parameter	Value
Goto tag	VEHICLE_STATUS
Icon display	Tag

"Goto" (Goto)

Table 2.11. "Goto" Parameters

Parameter	Value
Tag	local_acc
Icon display	Tag
Tag visibility	global

"Goto1" (Goto)

Table 2.12. "Goto1" Parameters

Parameter	Value
Tag	local_gps
Icon display	Tag
Tag visibility	global

"Goto2" (Goto)

Table 2.13. "Goto2" Parameters

Parameter	Value
Tag	FOREIGN_GPS
Icon display	Tag
Tag visibility	global

"Goto3" (Goto)

Table 2.14. "Goto3" Parameters

Parameter	Value
Tag	foreign_receiving_flag
Icon display	Tag
Tag visibility	global

"Goto4" (Goto)

Table 2.15. "Goto4" Parameters

Parameter	Value
Tag	DELTA
Icon display	Tag
Tag visibility	global

"Goto5" (Goto)

Table 2.16. "Goto5" Parameters

Parameter	Value
Tag	acc_offset
Icon display	Tag
Tag visibility	global

"Goto6" (Goto)

Table 2.17. "Goto6" Parameters

Parameter	Value
Tag	acc_calibrated
Icon display	Tag
Tag visibility	global

"Goto7" (Goto)

Table 2.18. "Goto7" Parameters

Parameter	Value
Tag	TRANSMIT
Icon display	Tag
Tag visibility	global

"Goto8" (Goto)

Table 2.19. "Goto8" Parameters

Parameter	Value
Tag	VEHICLE_STATUS
Icon display	Tag
Tag visibility	global

"Goto9" (Goto)

Table 2.20. "Goto9" Parameters

Parameter	Value
Tag	local_gyro
Icon display	Tag
Tag visibility	global

"GPS_PARSER" (MATLABSystem)

Table 2.21. "GPS_PARSER" Parameters

Parameter	Value
Sample time	TS_GPS

"Signal Conversion" (SignalConversion)

Table 2.22. "Signal Conversion" Parameters

Parameter	Value
Output	Nonvirtual bus
Data type	Bus: accel
Exclude this block from 'Block reduction' optimization	off

"Signal Conversion1" (SignalConversion)

Table 2.23. "Signal Conversion1" Parameters

Parameter	Value
Output	Nonvirtual bus
Data type	Bus: accel
Exclude this block from 'Block reduction' optimization	off

Block Execution Order

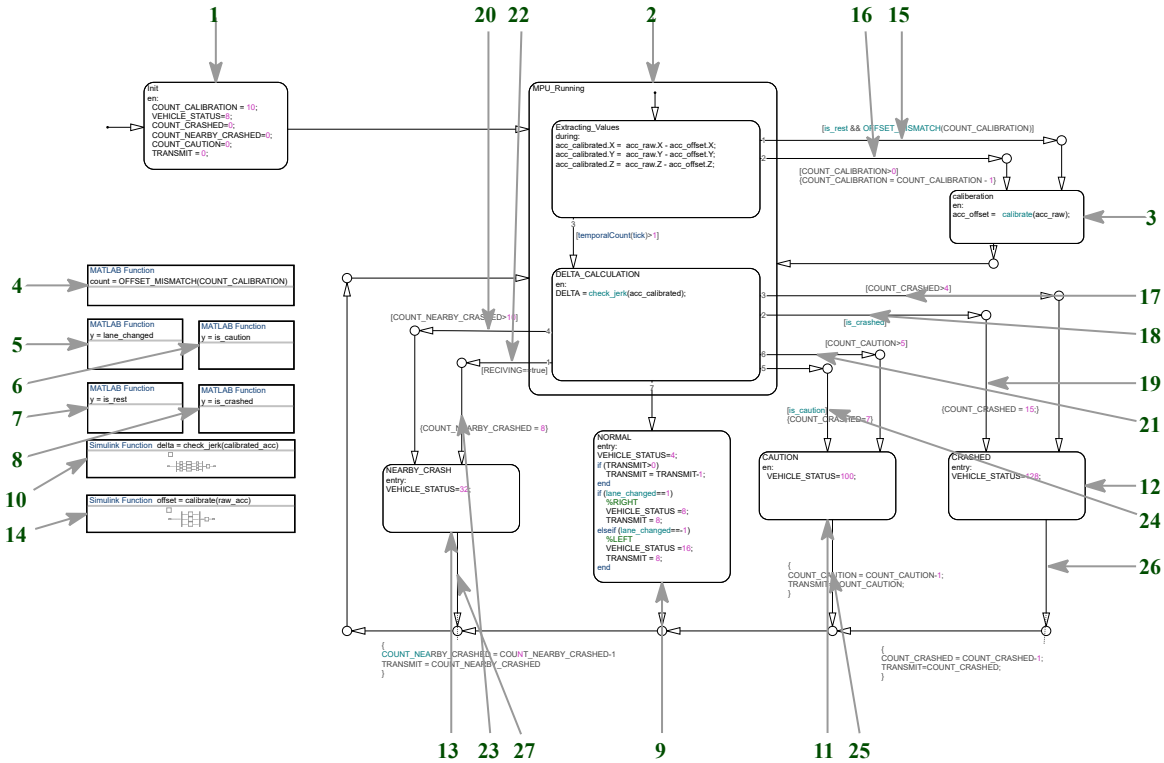
1. Constant1 (Constant)
2. *TRANSMITTER_READING*
    1. Constant (Constant)
    2. Constant1 (Constant)
    3. Unit Delay (UnitDelay)
    4. *MATLAB Function*
        1. SFunction (S-Function)
    5. Relational Operator (RelationalOperator)
    6. Sum of Elements (Sum)
    7. Cast To Boolean (DataTypeConversion)
    8. Data Type Conversion1 (DataTypeConversion)
    9. Data Type Conversion2 (DataTypeConversion)
    10. Data Type Conversion3 (DataTypeConversion)
    11. Data Type Conversion4 (DataTypeConversion)

    12. I2C Read (MATLABSystem)
    13. IsNaN (RelationalOperator)
    14. IsNaN1 (RelationalOperator)
    15. IsNaN2 (RelationalOperator)
    16. Switch (Switch)
    17. Switch1 (Switch)
    18. Switch2 (Switch)
    19. Signal Conversion_BusCreator1 (BusCreator)
3. *MPU6050*
    1. SFunction (S-Function)
    2. *i2cRd*
        1. I2C Read (MATLABSystem)
        2. Data Type Conversion (DataTypeConversion)
    3. *i2cWr*
        1. Ecriture I2C (MATLABSystem)
4. Signal Conversion_BusCreator1 (BusCreator)
5. *CORE_LOGIC*
    1. SFunction (S-Function)
    2. *check_jerk*
        1. Delay (Delay)
        2. Delay1 (Delay)
        3. Delay2 (Delay)
        4. UD (UnitDelay)
        5. Diff (Sum)
        6. UD (UnitDelay)
        7. Diff (Sum)
        8. UD (UnitDelay)
        9. Diff (Sum)
        10. Quantizer (Quantizer)
        11. Quantizer1 (Quantizer)
        12. Quantizer2 (Quantizer)
        13. Signal Conversion1_BusCreator1 (BusCreator)
    3. *calibrate*
        1. Moving Average (MATLABSystem)
        2. Moving Average1 (MATLABSystem)
        3. Moving Average2 (MATLABSystem)
        4. Signal Conversion_BusCreator1 (BusCreator)
6. Cast To Boolean (DataTypeConversion)
7. GPS_PARSER (MATLABSystem)
8. HiddenBuf_InsertedFor_TRANSMITTER_WRITTING_at_inport_3 (SignalConversion)
9. Signal Conversion1_BusCreator1 (BusCreator)
10. *TRANSMITTER_WRITTING*
    1. *MATLAB Function*
        1. SFunction (S-Function)
    2. I2C Write (MATLABSystem)
    3. Data Type Conversion (DataTypeConversion)
11. *MATLAB Function*
    1. SFunction (S-Function)

12. Serial Transmit (MATLABSystem)
13. SigConversion_InsertedFor_Bus Selector2_at_outport_0 (SignalConversion)
14. SigConversion_InsertedFor_Bus Selector2_at_outport_1 (SignalConversion)
15. SigConversion_InsertedFor_Bus Selector2_at_outport_2 (SignalConversion)
16. SigConversion_InsertedFor_Bus Selector_at_outport_0 (SignalConversion)
17. SigConversion_InsertedFor_Bus Selector_at_outport_1 (SignalConversion)
18. SigConversion_InsertedFor_Bus Selector_at_outport_2 (SignalConversion)
19. SigConversion_InsertedFor_Bus Selector6_at_outport_0 (SignalConversion)
20. SigConversion_InsertedFor_Bus Selector6_at_outport_1 (SignalConversion)
21. SigConversion_InsertedFor_Bus Selector6_at_outport_2 (SignalConversion)
22. Scope (Scope)
23. Display3 (Display)
24. Display4 (Display)
25. Display5 (Display)
26. SigConversion_InsertedFor_Bus Selector8_at_outport_0 (SignalConversion)
27. SigConversion_InsertedFor_Bus Selector9_at_outport_0 (SignalConversion)
28. SigConversion_InsertedFor_Bus Selector8_at_outport_1 (SignalConversion)
29. SigConversion_InsertedFor_Bus Selector9_at_outport_1 (SignalConversion)
30. SigConversion_InsertedFor_Bus Selector8_at_outport_2 (SignalConversion)
31. SigConversion_InsertedFor_Bus Selector9_at_outport_2 (SignalConversion)
32. SigConversion_InsertedFor_Bus Selector7_at_outport_0 (SignalConversion)
33. SigConversion_InsertedFor_Bus Selector7_at_outport_1 (SignalConversion)
34. SigConversion_InsertedFor_Bus Selector7_at_outport_2 (SignalConversion)
35. SigConversion_InsertedFor_Bus Selector10_at_outport_0 (SignalConversion)
36. SigConversion_InsertedFor_Bus Selector10_at_outport_1 (SignalConversion)
37. SigConversion_InsertedFor_Bus Selector10_at_outport_2 (SignalConversion)
38. Scope1 (Scope)
39. TAQSigLogging_InsertedFor_acc_calibrated_at_outport_0 (ToAsyncQueueBlock)
40. TAQSigLogging_InsertedFor_acc_offset_at_outport_0 (ToAsyncQueueBlock)
41. TAQSigLogging_InsertedFor_delta_at_outport_0 (ToAsyncQueueBlock)
42. TAQSigLogging_InsertedFor_local_acc_at_outport_0 (ToAsyncQueueBlock)
43. SigConversion_InsertedFor_Bus Selector5_at_outport_2 (SignalConversion)
44. Display1 (Display)
45. SigConversion_InsertedFor_Bus Selector5_at_outport_0 (SignalConversion)
46. Display2 (Display)
47. SigConversion_InsertedFor_Bus Selector5_at_outport_1 (SignalConversion)
48. Display22 (Display)
49. SigConversion_InsertedFor_Bus Selector5_at_outport_3 (SignalConversion)
50. Display23 (Display)
51. SigConversion_InsertedFor_Bus Selector5_at_outport_4 (SignalConversion)
52. Display24 (Display)

State Charts

Chart



1. Init
2. MPU_Running
3. caliberation
4. OFFSET_MISMATCH
5. lane_changed
6. is_caution
7. is_rest
8. is_crashed
9. NORMAL
10. check_jerk
11. CAUTION
12. CRASHED
13. NEARBY_CRASH
14. calibrate
15. [is_rest && OFFSET_MISMATCH(COUNT_CALIBRATION)]
16. [COUNT_CALIBRATION>0]...
17. [COUNT_CRASHED>4]
18. [is_crashed]
19. {COUNT_CRASHED = 15;}
20. [COUNT_NEARBY_CRASHED>10]
21. [COUNT_CAUTION>5]
22. [RECIVING==true]
23. {COUNT_NEARBY_CRASHED = 8}
24. [is_caution]...

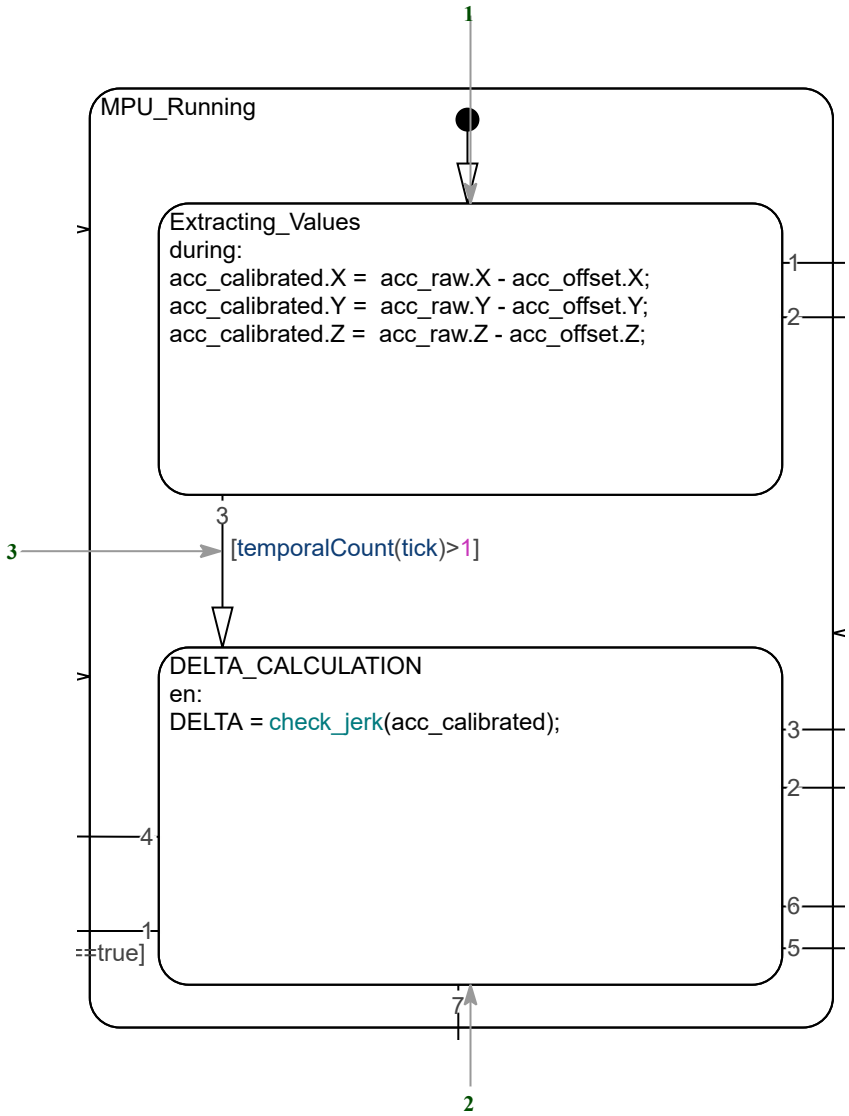
25. {...
26. {...
27. {...

States

OR State - Init

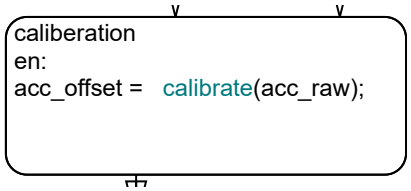
```
Init
en:
  COUNT_CALIBRATION = 10;
  VEHICLE_STATUS=8;
  COUNT_CRASHED=0;
  COUNT_NEARBY_CRASHED=0;
  COUNT_CAUTION=0;
  TRANSMIT = 0;
```

OR State - MPU_Running



1. Extracting_Values
2. DELTA_CALCULATION
3. [temporalCount(tick)>1]

OR State - caliberation



EMFunction - OFFSET_MISMATCH

MATLAB Function
count = OFFSET_MISMATCH(COUNT_CALIBRATION)

Table 2.24. OFFSET_MISMATCH Argument Summary

Name	Scope	Port	Data Type	Size
COUNT_CALIBRATION	Input	NaN	unknown	
count	Output	NaN	unknown	

OFFSET_MISMATCH Function Script

```
function count = OFFSET_MISMATCH(COUNT_CALIBRATION)

count = COUNT_CALIBRATION;
if(abs(acc_raw.X - acc_offset.X) > 0.01 && ...
   abs(acc_raw.Y - acc_offset.Y) > 0.01 && ...
   abs(acc_raw.Z - acc_offset.Z) > 0.01)

    count = COUNT_CALIBRATION +1;
end
```

EMFunction - lane_changed

MATLAB Function
y = lane_changed

Table 2.25. lane_changed Argument Summary

Name	Scope	Port	Data Type	Size
y	Output	NaN	unknown	

lane_changed Function Script

```
function y = lane_changed

y = 0;

if (acc_calibrated.X > THRESHOLD_LANE_CHANGE)
    % LEFT
    y = 1;
elseif (acc_calibrated.X < -THRESHOLD_LANE_CHANGE)
```

```
    % RIGHT
    y = -1;
end
```

EMFunction - is_caution

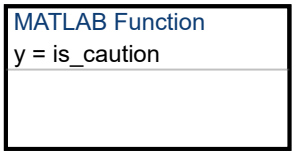


Table 2.26. is_caution Argument Summary

Name	Scope	Port	Data Type	Size
y	Output	NaN	unknown	

is_caution Function Script

```
function y = is_caution


if (abs(DELTA.Y) >= THRESHOLD_CAUTION)

    y = true;
else
    y = false;
end
```

EMFunction - is_rest

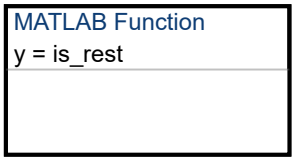


Table 2.27. is_rest Argument Summary

Name	Scope	Port	Data Type	Size
y	Output	NaN	unknown	

is_rest Function Script

```
function y = is_rest

if (abs(DELTA.X) >= THRESHOLD_REST &&...
    abs(DELTA.Y) >= THRESHOLD_REST &&...
    abs(DELTA.Z) >= THRESHOLD_REST )
```

```
    y = true;
else
    y = false;
end
```

EMFunction - is_crashed

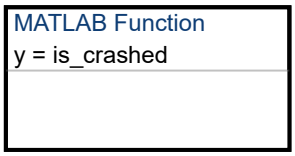


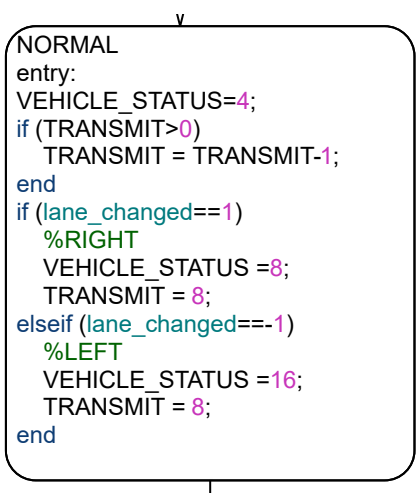
Table 2.28. is_crashed Argument Summary

Name	Scope	Port	Data Type	Size
y	Output	NaN	unknown	

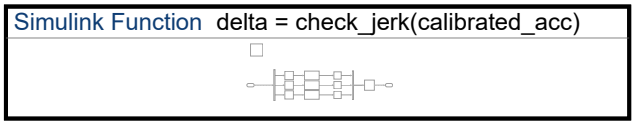
is_crashed Function Script

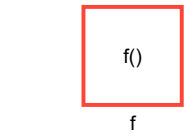
```
function y = is_crashed

if (abs(DELTA.Y) >= THRESHOLD_CRASHED)
    y = true;
else
    y = false;
end
```

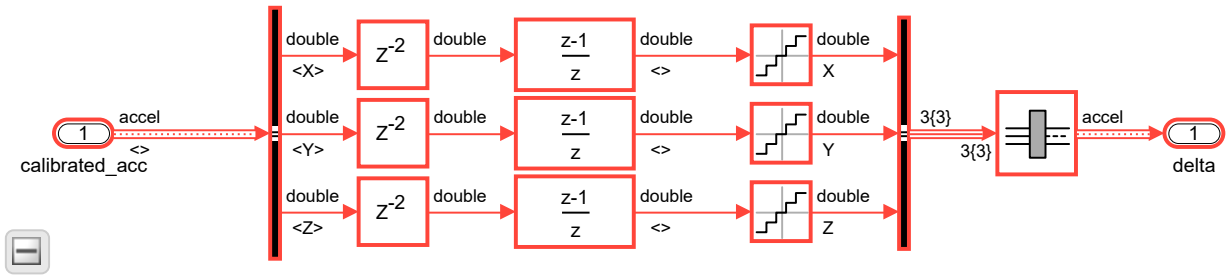
OR State - NORMAL

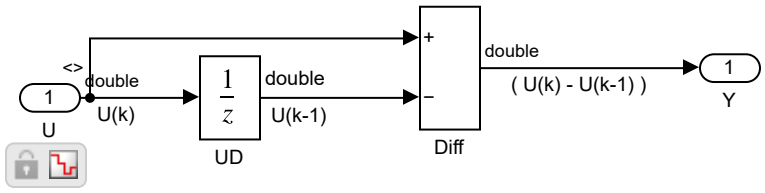


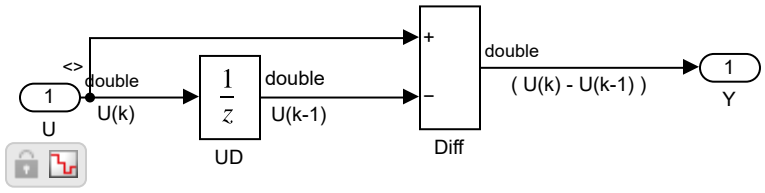
SLFunction - check_jerk

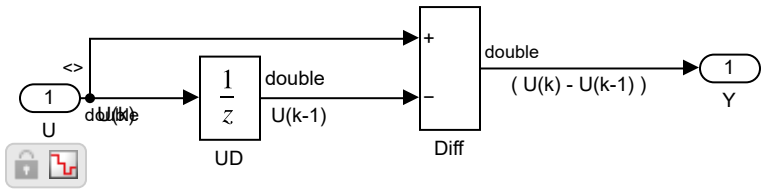












OR State - CAUTION

```
CAUTION
en:
  VEHICLE_STATUS=100;
```

OR State - CRASHED

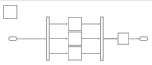
```
CRASHED
entry:
VEHICLE_STATUS=128;
```

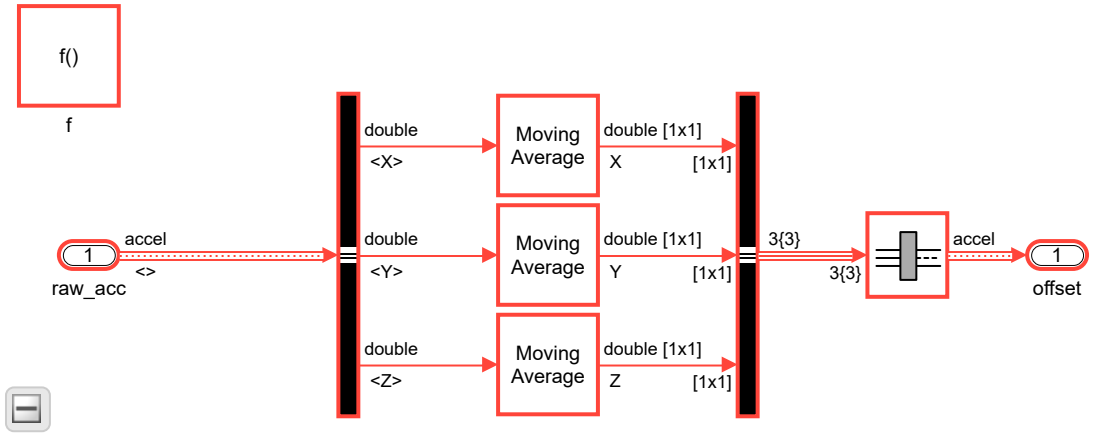
OR State - NEARBY_CRASH

```
NEARBY_CRASH
entry:
VEHICLE_STATUS=32;
```

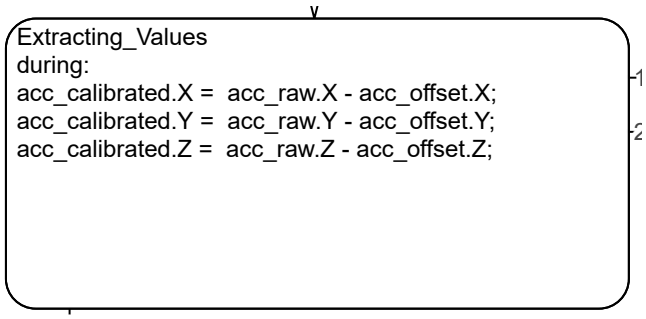
SLFunction - calibrate

```
Simulink Function  offset = calibrate(raw_acc)
```

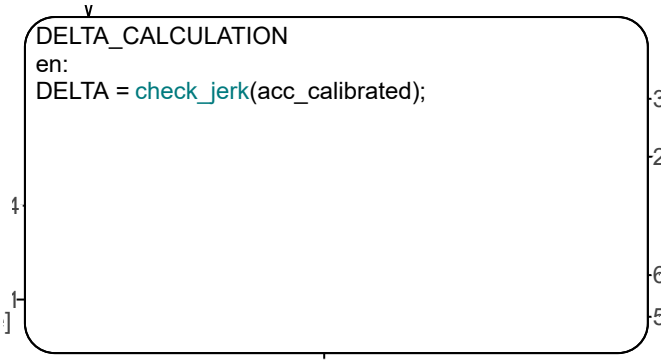




OR State - Extracting_Values



OR State - DELTA_CALCULATION



Data

Table 2.29. Data - acc_calibrated

Scope	Output
Data Type	Bus: accel
InitValue	struct('X', 0, 'Y', 0, 'Z', 0)

Table 2.30. Data - acc_offset

Scope	Output
Data Type	Bus: accel
InitValue	struct('X', 0, 'Y', 0, 'Z', 0)

Table 2.31. Data - acc_raw

Scope	Input
Data Type	Bus: accel

Table 2.32. Data - calibrated_acc

Scope	Input
Data Type	Bus: accel

Table 2.33. Data - count

Scope	Output
Data Type	Inherit: From definition in chart

Table 2.34. Data - COUNT_CALIBRATION

Scope	Local
Data Type	uint8

Table 2.35. Data - COUNT_CALIBRATION

Scope	Input
Data Type	Inherit: From definition in chart

Table 2.36. Data - COUNT_CAUTION

Scope	Local
Data Type	uint8

Table 2.37. Data - COUNT_CRASHED

Scope	Local
Data Type	uint8

Table 2.38. Data - COUNT_NEARBY_CRASHED

Scope	Local

Data Type	uint8

Table 2.39. Data - DELTA

Scope	Output
Data Type	Bus: accel
InitValue	struct('X', 0, 'Y', 0, 'Z', 0)

Table 2.40. Data - delta

Scope	Output
Data Type	Bus: accel

Table 2.41. Data - offset

Scope	Output
Data Type	Bus: accel

Table 2.42. Data - raw_acc

Scope	Input
Data Type	Bus: accel

Table 2.43. Data - RECIVING

Scope	Input
Data Type	boolean

Table 2.44. Data - THRESHOLD_CAUTION

Scope	Parameter
Data Type	Inherit: Same as Simulink

Table 2.45. Data - THRESHOLD_CRASHED

Scope	Parameter
Data Type	Inherit: Same as Simulink

Table 2.46. Data - THRESHOLD_LANE_CHANGE

Scope	Parameter
Data Type	Inherit: Same as Simulink

Table 2.47. Data - THRESHOLD_REST

Scope	Parameter
Data Type	Inherit: Same as Simulink

Table 2.48. Data - TRANSMIT

Scope	Output
Data Type	uint8

Table 2.49. Data - VEHICLE_STATUS

Scope	Output
Data Type	Inherit: Same as Simulink

Table 2.50. Data - y

Scope	Output
Data Type	Inherit: From definition in chart

Table 2.51. Data - y

Scope	Output
Data Type	Inherit: From definition in chart

Table 2.52. Data - y

Scope	Output
Data Type	Inherit: From definition in chart

Table 2.53. Data - y

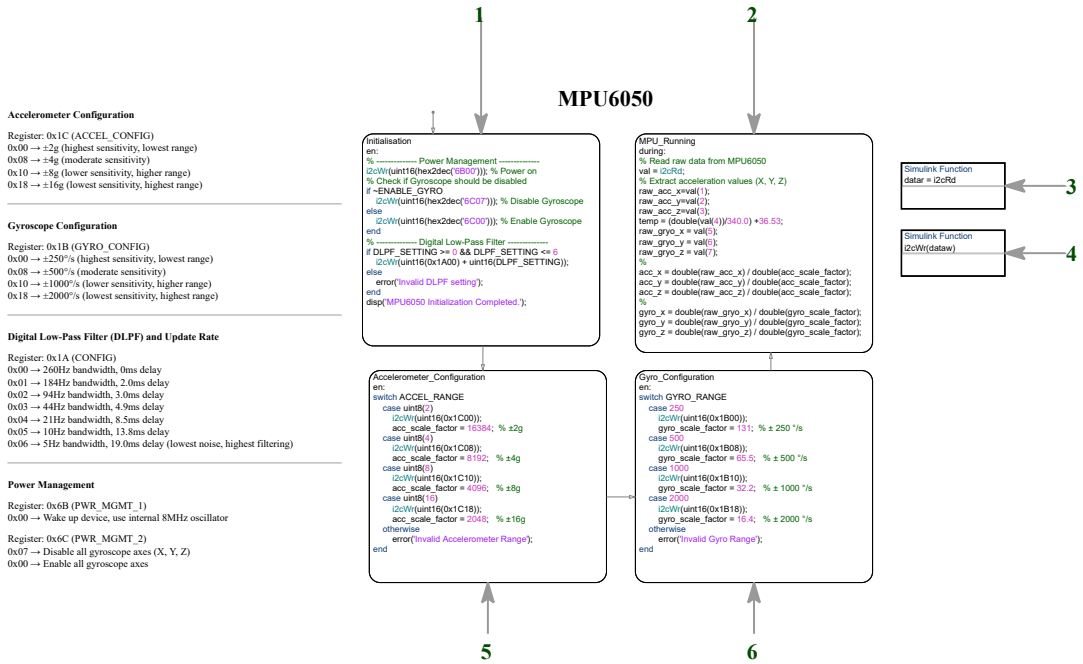
Scope	Output
Data Type	Inherit: From definition in chart

Embedded MATLAB Supporting Functions

Table 2.54. CORE_LOGIC Supporting Functions

Function	Defined By	Path
abs	MATLAB	

Chart



1. Initialisation
2. MPU_Running
3. i2cRd
4. i2cWr
5. Accelerometer_Configuration
6. Gyro_Configuration

States

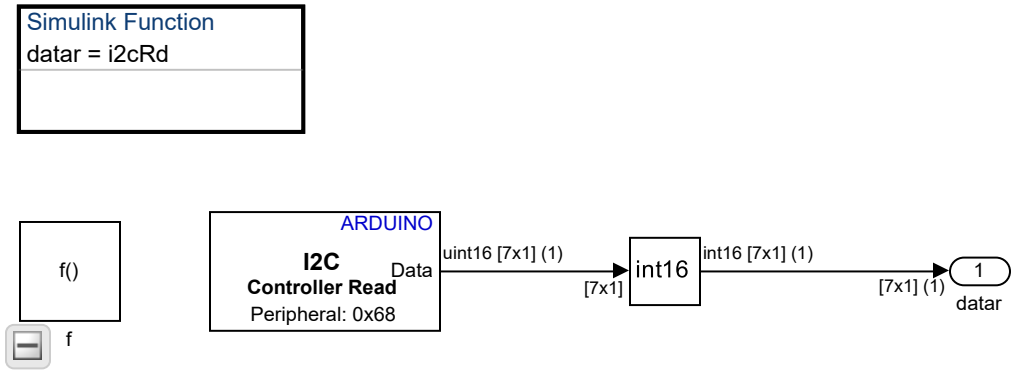
OR State - Initialisation

```
Initialisation
en:
% -------------- Power Management --------------
i2cWr(uint16(hex2dec('6B00'))); % Power on
% Check if Gyroscope should be disabled
if ~ENABLE_GYRO
   i2cWr(uint16(hex2dec('6C07'))); % Disable Gyroscope
else
   i2cWr(uint16(hex2dec('6C00'))); % Enable Gyroscope
end
% -------------- Digital Low-Pass Filter --------------
if DLPF_SETTING >= 0 && DLPF_SETTING <= 6
   i2cWr(uint16(0x1A00) + uint16(DLPF_SETTING));
else
   error('Invalid DLPF setting');
end
disp('MPU6050 Initialization Completed.');
```

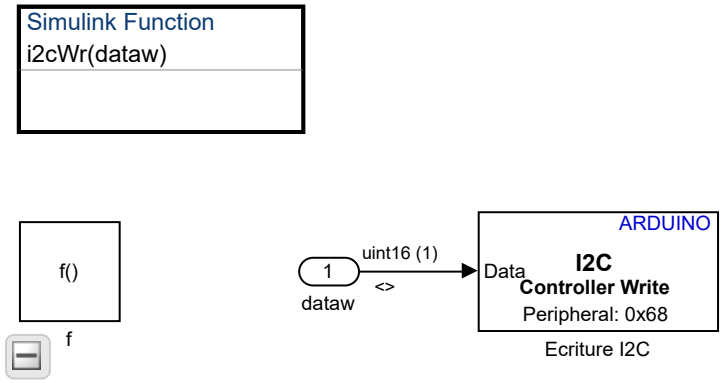
OR State - MPU_Running

```
MPU_Running
during:
% Read raw data from MPU6050
val = i2cRd;
% Extract acceleration values (X, Y, Z)
raw_acc_x=val(1);
raw_acc_y=val(2);
raw_acc_z=val(3);
temp = (double(val(4))/340.0) +36.53;
raw_gryo_x = val(5);
raw_gryo_y = val(6);
raw_gryo_z = val(7);
%
acc_x = double(raw_acc_x) / double(acc_scale_factor);
acc_y = double(raw_acc_y) / double(acc_scale_factor);
acc_z = double(raw_acc_z) / double(acc_scale_factor);
%
gyro_x = double(raw_gryo_x) / double(gyro_scale_factor);
gyro_y = double(raw_gryo_y) / double(gyro_scale_factor);
gyro_z = double(raw_gryo_z) / double(gyro_scale_factor);
```

SLFunction - i2cRd



SLFunction - i2cWr



OR State - Accelerometer_Configuration

```
Accelerometer_Configuration
en:
switch ACCEL_RANGE
   case uint8(2)
      i2cWr(uint16(0x1C00));
      acc_scale_factor = 16384;  % ±2g
   case uint8(4)
      i2cWr(uint16(0x1C08));
      acc_scale_factor = 8192;   % ±4g
   case uint8(8)
      i2cWr(uint16(0x1C10));
      acc_scale_factor = 4096;   % ±8g
   case uint8(16)
      i2cWr(uint16(0x1C18));
      acc_scale_factor = 2048;   % ±16g
   otherwise
      error('Invalid Accelerometer Range');
end
```

OR State - Gyro_Configuration

```
Gyro_Configuration
en:
switch GYRO_RANGE
   case 250
      i2cWr(uint16(0x1B00));
      gyro_scale_factor = 131;  % ± 250 °/s
   case 500
      i2cWr(uint16(0x1B08));
      gyro_scale_factor = 65.5;   % ± 500 °/s
   case 1000
      i2cWr(uint16(0x1B10));
      gyro_scale_factor = 32.2;   % ± 1000 °/s
   case 2000
      i2cWr(uint16(0x1B18));
      gyro_scale_factor = 16.4;   % ± 2000 °/s
   otherwise
      error('Invalid Gyro Range');
end
```

Data

Table 2.55. Data - acc_scale_factor

Scope	Local
Data Type	Inherit: From definition in chart

Table 2.56. Data - acc_x

Scope	Output

Table 2.57. Data - acc_y

Scope	Output

Table 2.58. Data - acc_z

Scope	Output

Table 2.59. Data - ACCEL_RANGE

Scope	Parameter
Data Type	uint8

Table 2.60. Data - datar

Scope	Output
Data Type	Inherit: Same as Simulink

Table 2.61. Data - dataw

Scope	Input
Data Type	uint16

Table 2.62. Data - DLPF_SETTING

Scope	Parameter

Table 2.63. Data - ENABLE_GYRO

Scope	Parameter

Table 2.64. Data - GYRO_RANGE

Scope	Parameter
Data Type	Inherit: Same as Simulink

Table 2.65. Data - gyro_scale_factor

Scope	Local

Table 2.66. Data - gyro_x

Scope	Output
Data Type	Inherit: Same as Simulink

Table 2.67. Data - gyro_y

Scope	Output
Data Type	Inherit: Same as Simulink

Table 2.68. Data - gyro_z

Scope	Output
Data Type	Inherit: Same as Simulink

Table 2.69. Data - raw_acc_x

Scope	Local

Data Type	int16

Table 2.70. Data - raw_acc_y

Scope	Local
Data Type	int16

Table 2.71. Data - raw_acc_z

Scope	Local
Data Type	int16

Table 2.72. Data - raw_gryo_x

Scope	Local
Data Type	Inherit: From definition in chart

Table 2.73. Data - raw_gryo_y

Scope	Local
Data Type	Inherit: From definition in chart

Table 2.74. Data - raw_gryo_z

Scope	Local
Data Type	Inherit: From definition in chart

Table 2.75. Data - temp

Scope	Local
Data Type	Inherit: From definition in chart

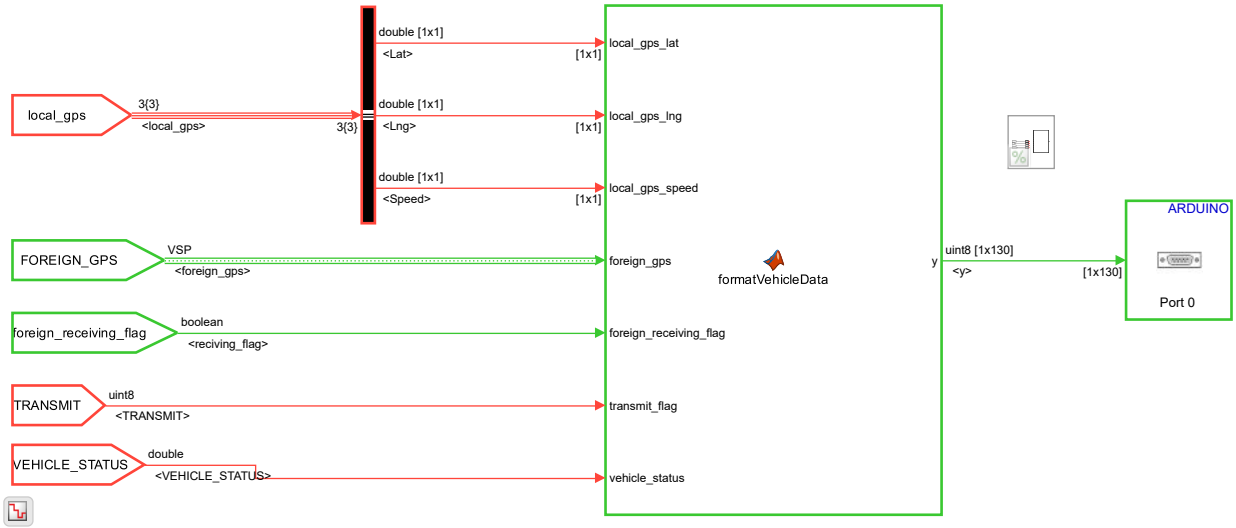
Table 2.76. Data - val

Scope	Local
Data Type	int16

Chapter 3. Subsystems

serial_printing

Figure 3.1. warning_system/ serial_printing



Blocks

Parameters

"Bus Selector" (BusSelector)

Table 3.1. "Bus Selector" Parameters

Parameter	Value
Output signals	Lat,Lng,Speed
Output as virtual bus	off
InputSignals	Lat Lng Speed

Output Hierarchy:

1. *Bus Selector*
    1. <Lat>
    2. <Lng>

3. <Speed>

"From2" (From)

Table 3.2. "From2" Parameters

Parameter	Value
Goto tag	local_gps
Icon display	Tag

"From3" (From)

Table 3.3. "From3" Parameters

Parameter	Value
Goto tag	TRANSMIT
Icon display	Tag

"From5" (From)

Table 3.4. "From5" Parameters

Parameter	Value
Goto tag	FOREIGN_GPS
Icon display	Tag

"From6" (From)

Table 3.5. "From6" Parameters

Parameter	Value
Goto tag	foreign_receiving_flag
Icon display	Tag

"From9" (From)

Table 3.6. "From9" Parameters

Parameter	Value
Goto tag	VEHICLE_STATUS
Icon display	Tag

"MATLAB Function" (MATLAB Function)

Table 3.7. MATLAB Function Function Properties

Property	Value
Update Method	INHERITED
Sample Time	-1
Support variable-size arrays	1
Saturate on integer overflow	1
Treat these inherited Simulink signal types as fi objects	Fixed-point
MATLAB Function block fimath	Same as MATLAB Default
Input fi math	fimath(...)
Description	

Table 3.8. MATLAB Function Argument Summary

Name	Scope	Port	Data Type	Size
y	Output	1	uint8	[1, 130]
local_gps_lat	Input	1	double	1
local_gps_lng	Input	2	double	1
local_gps_speed	Input	3	double	1
foreign_gps	Input	4	VSP	1
foreign_receiving_flag	Input	5	boolean	1
transmit_flag	Input	6	uint8	1
vehicle_status	Input	7	double	1

MATLAB Function Function Script

```
function y = formatVehicleData( ...
    local_gps_lat, local_gps_lng, local_gps_speed, foreign_gps, ...
    foreign_receiving_flag, transmit_flag, ...
    vehicle_status)
    %#codegen

    % Pre-allocate fixed-size output (600 bytes)
    y = zeros(1, 130, 'uint8');  % fixed 600-byte output
    str = blanks(130);           % pre-fill with 600 spaces (char array)
```

```
    % Force logicals to int
    transmit_flag = int32(transmit_flag);
    foreign_receiving_flag = int32(foreign_receiving_flag);
    vehicle_status = int32(vehicle_status);

    switch vehicle_status
        case 0
            statusStr = 'INVALID';
        case 4
            statusStr = 'NORMAL';
        case 8
            statusStr = 'LANE_CHANGE_RIGHT';
        case 16
            statusStr = 'LANE_CHANGE_LEFT';
        case 32
            statusStr = 'NEARBY_CRASH';
        case 128
            statusStr = 'CRASH';
        case 100
            statusStr = 'CAUTION';
        otherwise
            statusStr = 'UNKNOWN';
    end

    % Format the message string
    formatted = sprintf([ '\n<%17s (%3d) || TRANSMIT: %2d || (%.7f, %.7f,
%.3f) || RECEIVING: %d || (%.7f, %.7f, %.3f)>'], ...
    statusStr, vehicle_status, ...
    transmit_flag, local_gps_lat, local_gps_lng, local_gps_speed, ...
    foreign_receiving_flag , foreign_gps.lat, foreign_gps.lng, for-
eign_gps.speed ...
    );
    % Copy characters into the fixed-length string
    len = min(length(formatted), 130);
    str(1:len) = formatted(1:len);

    % Convert to uint8 for output
    y = uint8(str);
end
```

Table 3.9. MATLAB Function Supporting Functions

Function	Defined By	Path
blanks	MATLAB	
char	MATLAB	
feval	MATLAB	
fieldnames	MATLAB	
isequal	MATLAB	

Function	Defined By	Path
ismatrix	MATLAB	
isnan	MATLAB	
isrow	MATLAB	
issparse	MATLAB	
isstring	MATLAB	
min	MATLAB	
parseParameterInputsCG	MATLAB	
sprintf	MATLAB	

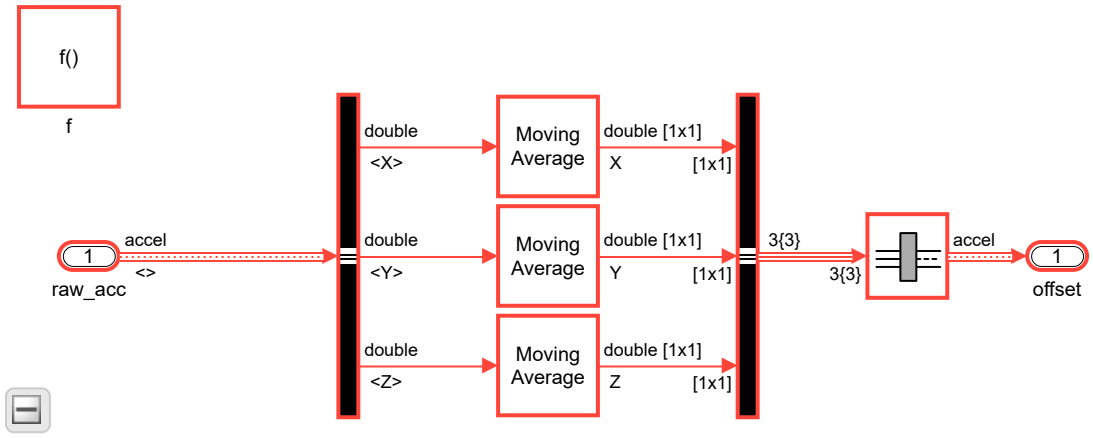
"Serial Transmit" (MATLABSystem)

Table 3.10. "Serial Transmit" Parameters

Parameter	Value
arduinoblockmaskSerialPortProp	0
arduinoblockmaskSerialSendModeProp	write
arduinoblockmaskSerialHeaderProp	[]
arduinoblockmaskSerialTerminatorProp	[]

calibrate

Figure 3.2. warning_system/CORE_LOGIC/calibrate



Blocks

Parameters

"Bus Creator" (BusCreator)

Table 3.11. "Bus Creator" Parameters

Parameter	Value
Number of inputs	3
Display option	bar
Data type	Bus: accel
Output as nonvirtual bus	off
Use names from inputs instead of from bus object	on

"Bus Selector" (BusSelector)

Table 3.12. "Bus Selector" Parameters

Parameter	Value
Output signals	X,Y,Z
Output as virtual bus	off
InputSignals	X Y Z

Output Hierarchy:

1. *Bus Selector*
    1. <X>
    2. <Y>
    3. <Z>

"f" (TriggerPort)

Table 3.13. "f" Parameters

Parameter	Value
Trigger type	function-call
Trigger time	on message available
Schedule as aperiodic partition	on
Treat as Simulink function	off
Execute function call asynchronously	off
Function visibility	global
Enable variant condition	off

Parameter	Value
Variant control	(inherit)
Generate preprocessor conditionals	off
States when enabling	held
Propagate sizes of variable-size signals	During execution
Show output port	off
Sample time type	triggered
Sample time	1
Enable zero-crossing detection	on
Initial trigger signal state	compatibility (no trigger on first evaluation)
Port dimensions	-1
Trigger signal sample time	-1
Minimum	[]
Maximum	[]
Data type	Inherit: auto
Interpolate data	on
FunctionPrototype	f()

"Moving Average" (MATLABSystem)

Table 3.14. "Moving Average" Parameters

Parameter	Value
dspStatisticsBlockDialogMethod_MP	Sliding window
dspStatisticsBlockDialogSpecifyWindowLength_MP	on
dspStatisticsBlockDialogWindowLength_MP	4
dspStatisticsBlockDialogOverlapLength_MP	3
dspblkprm_promptsAllowArbitraryFixedInput_MP	off
SystemBlockMATLABSystemSimulateUsing	Code generation

"Moving Average1" (MATLABSystem)

Table 3.15. "Moving Average1" Parameters

Parameter	Value
dspStatisticsBlockDialogMethod_MP	Sliding window
dspStatisticsBlockDialogSpecifyWindowLength_MP	on
dspStatisticsBlockDialogWindowLength_MP	4

Parameter	Value
dspStatisticsBlockDialogOverlapLength_MP	3
dspblkprm_promptsAllowArbitraryFixedInput_MP	off
SystemBlockMATLABSystemSimulateUsing	Code generation

"Moving Average2" (MATLABSystem)

Table 3.16. "Moving Average2" Parameters

Parameter	Value
dspStatisticsBlockDialogMethod_MP	Sliding window
dspStatisticsBlockDialogSpecifyWindowLength_MP	on
dspStatisticsBlockDialogWindowLength_MP	4
dspStatisticsBlockDialogOverlapLength_MP	3
dspblkprm_promptsAllowArbitraryFixedInput_MP	off
SystemBlockMATLABSystemSimulateUsing	Code generation

"offset" (Outport)

Table 3.17. "offset" Parameters

Parameter	Value
Port number	1
Icon display	Port number
Output function call	off
Minimum	[]
Maximum	[]
Data type	Bus: accel
Lock output data type setting against changes by the fixed-point tools	off
Output as nonvirtual bus in parent model	off
Bus virtuality	inherit
Data mode	inherit
Unit (e.g., m, m/s^2, N*m)	inherit
Port dimensions (-1 for inherited)	-1
Variable-size signal	No
Sample time (-1 for inherited)	-1
Signal type	real
Ensure outport is virtual	off

Parameter	Value
Output when disabled	held
Initial output	[]
MustResolveToSignalObject	off
Specify output when source is unconnected	off
Constant value	0
Interpret vector parameters as 1-D	on

"raw_acc" (Inport)

Table 3.18. "raw_acc" Parameters

Parameter	Value
Port number	1
Port dimensions (-1 for inherited)	1
Sample time (-1 for inherited)	-1
Minimum	[]
Maximum	[]
Data type	Bus: accel

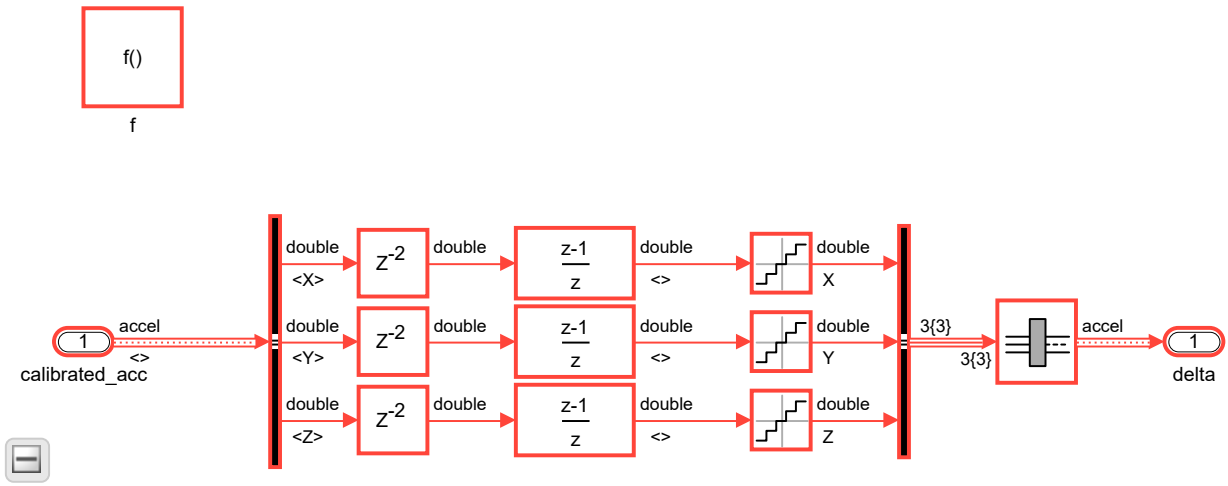
"Signal Conversion" (SignalConversion)

Table 3.19. "Signal Conversion" Parameters

Parameter	Value
Output	Nonvirtual bus
Data type	Bus: accel
Exclude this block from 'Block reduction' optimization	off

check_jerk

Figure 3.3. warning_system/CORE_LOGIC/check_jerk



Blocks

Parameters

"Bus Creator" (BusCreator)

Table 3.20. "Bus Creator" Parameters

Parameter	Value
Number of inputs	3
Display option	bar
Data type	Bus: accel
Output as nonvirtual bus	off
Use names from inputs instead of from bus object	on

"Bus Selector" (BusSelector)

Table 3.21. "Bus Selector" Parameters

Parameter	Value
Output signals	X,Y,Z
Output as virtual bus	off

Parameter	Value
InputSignals	X Y Z

Output Hierarchy:

1. *Bus Selector*
    1. <X>
    2. <Y>
    3. <Z>

"calibrated_acc" (Inport)

Table 3.22. "calibrated_acc" Parameters

Parameter	Value
Port number	1
Port dimensions (-1 for inherited)	1
Sample time (-1 for inherited)	-1
Minimum	[]
Maximum	[]
Data type	Bus: accel

"Delay" (Delay)

Table 3.23. "Delay" Parameters

Parameter	Value
Delay length source	Dialog
Delay length	2
Delay upper limit	100
Initial condition source	Dialog
Initial condition	0.0
External reset	None
Show enable port	off
Prevent direct feedthrough	off
Diagnostic for delay length	None
Remove delay length check in generated code	off
Input processing	Elements as channels (sample based)

Parameter	Value
Use circular buffer for state	off
Sample time (-1 for inherited)	-1
State name must resolve to Simulink signal object	off

"Delay1" (Delay)

Table 3.24. "Delay1" Parameters

Parameter	Value
Delay length source	Dialog
Delay length	2
Delay upper limit	100
Initial condition source	Dialog
Initial condition	0.0
External reset	None
Show enable port	off
Prevent direct feedthrough	off
Diagnostic for delay length	None
Remove delay length check in generated code	off
Input processing	Elements as channels (sample based)
Use circular buffer for state	off
Sample time (-1 for inherited)	-1
State name must resolve to Simulink signal object	off

"Delay2" (Delay)

Table 3.25. "Delay2" Parameters

Parameter	Value
Delay length source	Dialog
Delay length	2
Delay upper limit	100
Initial condition source	Dialog
Initial condition	0.0
External reset	None
Show enable port	off
Prevent direct feedthrough	off

Parameter	Value
Diagnostic for delay length	None
Remove delay length check in generated code	off
Input processing	Elements as channels (sample based)
Use circular buffer for state	off
Sample time (-1 for inherited)	-1
State name must resolve to Simulink signal object	off

"delta" (Outport)

Table 3.26. "delta" Parameters

Parameter	Value
Port number	1
Icon display	Port number
Output function call	off
Minimum	[]
Maximum	[]
Data type	Bus: accel
Lock output data type setting against changes by the fixed-point tools	off
Output as nonvirtual bus in parent model	off
Bus virtuality	inherit
Data mode	inherit
Unit (e.g., m, m/s^2, N*m)	inherit
Port dimensions (-1 for inherited)	-1
Variable-size signal	No
Sample time (-1 for inherited)	-1
Signal type	real
Ensure outport is virtual	off
Output when disabled	held
Initial output	[]
MustResolveToSignalObject	off
Specify output when source is unconnected	off
Constant value	0
Interpret vector parameters as 1-D	on

"Difference" (SubSystem)

Table 3.27. "Difference" Parameters

Parameter	Value
SimulinkmasksInitialConditionForPreviousInput_MP	0.0
SimulinkmasksInputProcessing_MP	Elements as channels (sample b ased)
SimulinkmasksOutputMinimum_MP	[]
SimulinkmasksOutputMaximum_MP	[]
SimulinkmasksOutputDataType_MP	Inherit: Inherit via internal rule
SimulinkmasksLockOutputDataTypeAgainstFxpTools_MP	off
SimulinkmasksIntegerRoundingMode_MP	Floor
SimulinkmasksSaturateToMaxOrMinWhenOverflowsOccur_MP	off

"Difference1" (SubSystem)

Table 3.28. "Difference1" Parameters

Parameter	Value
SimulinkmasksInitialConditionForPreviousInput_MP	0.0
SimulinkmasksInputProcessing_MP	Elements as channels (sample b ased)
SimulinkmasksOutputMinimum_MP	[]
SimulinkmasksOutputMaximum_MP	[]
SimulinkmasksOutputDataType_MP	Inherit: Inherit via internal rule
SimulinkmasksLockOutputDataTypeAgainstFxpTools_MP	off
SimulinkmasksIntegerRoundingMode_MP	Floor
SimulinkmasksSaturateToMaxOrMinWhenOverflowsOccur_MP	off

"Difference2" (SubSystem)

Table 3.29. "Difference2" Parameters

Parameter	Value
SimulinkmasksInitialConditionForPreviousInput_MP	0.0
SimulinkmasksInputProcessing_MP	Elements as channels (sample b ased)
SimulinkmasksOutputMinimum_MP	[]
SimulinkmasksOutputMaximum_MP	[]

Parameter	Value
SimulinkmasksOutputDataType_MP	Inherit: Inherit via internal rule
SimulinkmasksLockOutputDataTypeAgainstFxpTools_MP	off
SimulinkmasksIntegerRoundingMode_MP	Floor
SimulinkmasksSaturateToMaxOrMinWhenOverflowsOccur_MP	off

"f" (TriggerPort)

Table 3.30. "f" Parameters

Parameter	Value
Trigger type	function-call
Trigger time	on message available
Schedule as aperiodic partition	on
Treat as Simulink function	off
Execute function call asynchronously	off
Function visibility	global
Enable variant condition	off
Variant control	(inherit)
Generate preprocessor conditionals	off
States when enabling	held
Propagate sizes of variable-size signals	During execution
Show output port	off
Sample time type	triggered
Sample time	1
Enable zero-crossing detection	on
Initial trigger signal state	compatibility (no trigger on first evaluation)
Port dimensions	-1
Trigger signal sample time	-1
Minimum	[]
Maximum	[]
Data type	Inherit: auto
Interpolate data	on
FunctionPrototype	f()

"Quantizer" (Quantizer)

Table 3.31. "Quantizer" Parameters

Parameter	Value
Quantization interval	0.1
Treat as gain when linearizing	on
Sample time (-1 for inherited)	-1

"Quantizer1" (Quantizer)

Table 3.32. "Quantizer1" Parameters

Parameter	Value
Quantization interval	0.1
Treat as gain when linearizing	on
Sample time (-1 for inherited)	-1

"Quantizer2" (Quantizer)

Table 3.33. "Quantizer2" Parameters

Parameter	Value
Quantization interval	0.1
Treat as gain when linearizing	on
Sample time (-1 for inherited)	-1

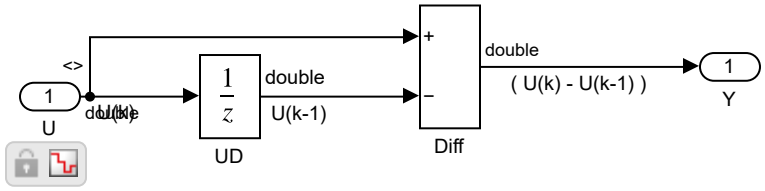
"Signal Conversion1" (SignalConversion)

Table 3.34. "Signal Conversion1" Parameters

Parameter	Value
Output	Nonvirtual bus
Data type	Bus: accel
Exclude this block from 'Block reduction' optimization	off

Difference

Figure 3.4. warning_system/CORE_LOGIC/check_jerk/Difference



Blocks

Parameters

"Diff" (Sum)

Table 3.35. "Diff" Parameters

Parameter	Value
Icon shape	rectangular
List of signs	+-
Apply over	All dimensions
Dimension	1
Output minimum	OutMin
Output maximum	OutMax
Output data type	OutDataTypeStr
Accumulator data type	Inherit: Inherit via internal rule
Require all inputs to have the same data type	off
Lock data type settings against changes by the fixed-point tools	off
Integer rounding mode	Floor
Saturate on integer overflow	off
Sample time (-1 for inherited)	-1

"U" (Inport)

Table 3.36. "U" Parameters

Parameter	Value
Port number	1

Parameter	Value
Port dimensions (-1 for inherited)	-1
Sample time (-1 for inherited)	-1
Minimum	[]
Maximum	[]
Data type	Inherit: auto

"UD" (UnitDelay)

Table 3.37. "UD" Parameters

Parameter	Value
Initial condition	ICPrevInput
Input processing	Elements as channels (sample based)
Sample time (-1 for inherited)	-1
State name must resolve to Simulink signal object	off

"Y" (Outport)

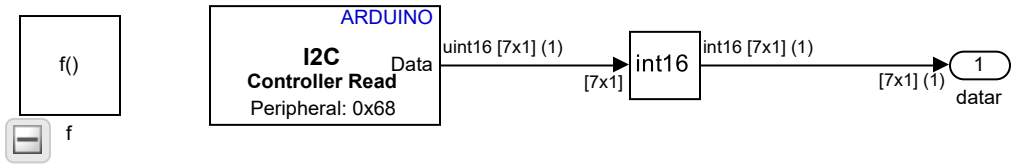
Table 3.38. "Y" Parameters

Parameter	Value
Port number	1
Icon display	Port number
Output function call	off
Minimum	OutMin
Maximum	OutMax
Data type	Inherit: auto
Lock output data type setting against changes by the fixed-point tools	off
Output as nonvirtual bus in parent model	off
Bus virtuality	inherit
Data mode	inherit
Unit (e.g., m, m/s^2, N*m)	inherit
Port dimensions (-1 for inherited)	-1
Variable-size signal	Inherit
Sample time (-1 for inherited)	-1
Ensure outport is virtual	off
Output when disabled	held

Parameter	Value
Initial output	0
MustResolveToSignalObject	off
Specify output when source is unconnected	off
Constant value	0
Interpret vector parameters as 1-D	on

i2cRd

Figure 3.5. warning_system/MPU6050/i2cRd



Blocks

Parameters

"Data Type Conversion" (DataTypeConversion)

Table 3.39. "Data Type Conversion" Parameters

Parameter	Value
Output minimum	[]
Output maximum	[]
Output data type	int16
Lock output data type setting against changes by the fixed-point tools	off
Input and output to have equal	Real World Value (RWV)
Integer rounding mode	Floor
Saturate on integer overflow	off
Sample time (-1 for inherited)	-1

"datar" (Outport)

Table 3.40. "datar" Parameters

Parameter	Value
Port number	1
Icon display	Port number
Output function call	off
Minimum	[]
Maximum	[]
Data type	Inherit: auto
Lock output data type setting against changes by the fixed-point tools	off
Output as nonvirtual bus in parent model	off
Bus virtuality	inherit
Data mode	inherit
Unit (e.g., m, m/s^2, N*m)	1
Port dimensions (-1 for inherited)	-1
Variable-size signal	No
Sample time (-1 for inherited)	-1
Signal type	real
Ensure outport is virtual	off
Output when disabled	held
Initial output	[]
MustResolveToSignalObject	off
Specify output when source is unconnected	off
Constant value	0
Interpret vector parameters as 1-D	off

"f" (TriggerPort)

Table 3.41. "f" Parameters

Parameter	Value
Trigger type	function-call
Trigger time	on message available
Schedule as aperiodic partition	on
Treat as Simulink function	off
Execute function call asynchronously	off

Parameter	Value
Function visibility	global
Enable variant condition	off
Variant control	Variant
Generate preprocessor conditionals	off
States when enabling	held
Propagate sizes of variable-size signals	During execution
Show output port	off
Sample time type	triggered
Sample time	inf
Enable zero-crossing detection	on
Initial trigger signal state	compatibility (no trigger on first evaluation)
Port dimensions	-1
Trigger signal sample time	-1
Minimum	[]
Maximum	[]
Data type	Inherit: auto
Interpolate data	on
FunctionPrototype	f()

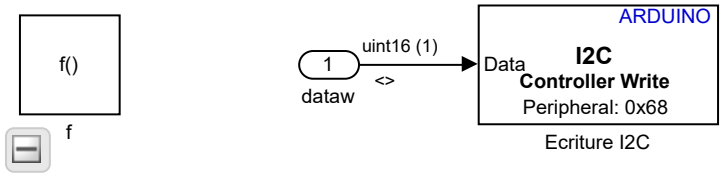
"I2C Read" (MATLABSystem)

Table 3.42. "I2C Read" Parameters

Parameter	Value
arduinoblockmaskI2CModuleProp	0
arduinoblockmaskSlaveAddressProp	hex2dec('68')
arduinoblockmaskSlaveByteOrderProp	BigEndian
arduinoblockmaskRegisterAddressModeProp	on
arduinoblockmaskRegisterAddressProp	hex2dec('3b')
arduinoblockmaskSlaveDataTypeProp	uint16
arduinoblockmaskDataLengthProp	7
arduinoblockmaskOutputStatusProp	off
arduinoblockmaskSampleTimeProp	-1

i2cWr

Figure 3.6. warning_system/MPU6050/i2cWr



Blocks

Parameters

"dataw" (Inport)

Table 3.43. "dataw" Parameters

Parameter	Value
Port number	1
Port dimensions (-1 for inherited)	1
Sample time (-1 for inherited)	-1
Minimum	[]
Maximum	[]
Data type	uint16

"Ecriture I2C" (MATLABSystem)

Table 3.44. "Ecriture I2C" Parameters

Parameter	Value
arduinoblockmaskI2CModuleProp	0
arduinoblockmaskSlaveAddressProp	hex2dec('68')
arduinoblockmaskSlaveByteOrderProp	BigEndian
arduinoblockmaskRegisterAddressModeProp	off
arduinoblockmaskRepeatedStartProp	off
arduinoblockmaskOutputStatusProp	off

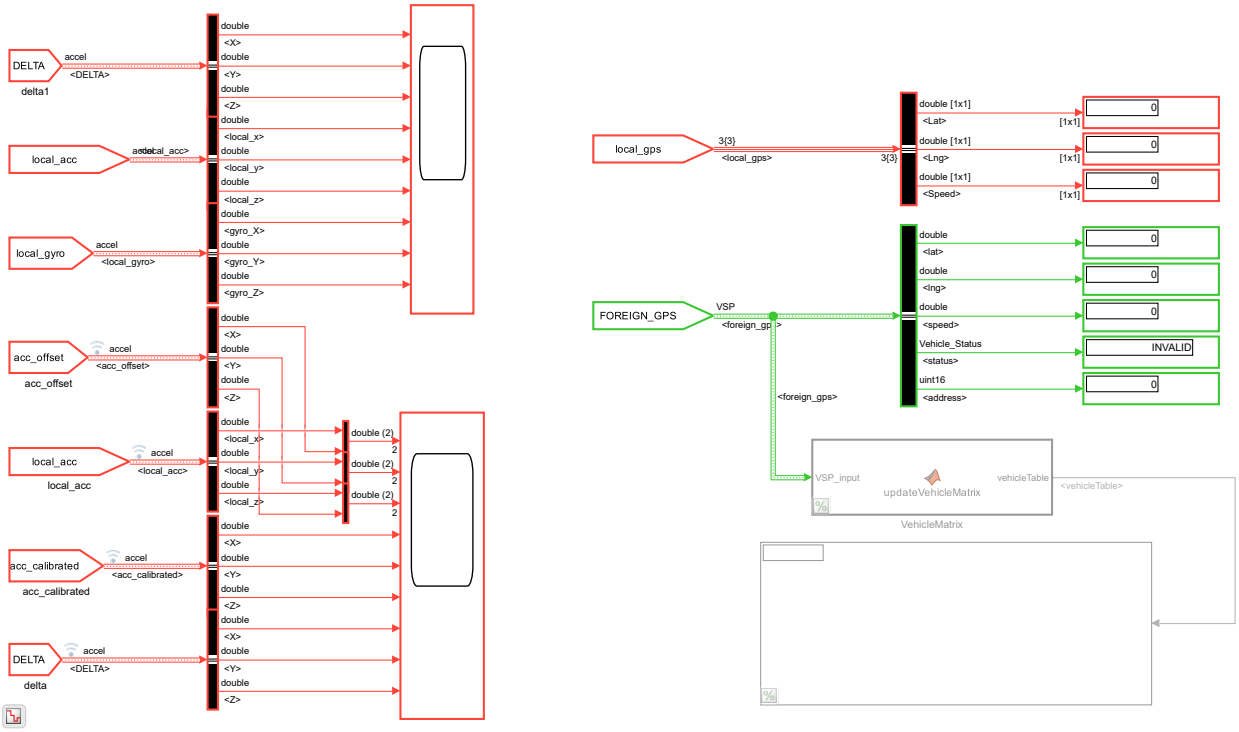
"f" (TriggerPort)

Table 3.45. "f" Parameters

Parameter	Value
Trigger type	function-call
Trigger time	on message available
Schedule as aperiodic partition	on
Treat as Simulink function	off
Execute function call asynchronously	off
Function visibility	global
Enable variant condition	off
Variant control	Variant
Generate preprocessor conditionals	off
States when enabling	held
Propagate sizes of variable-size signals	During execution
Show output port	off
Sample time type	triggered
Sample time	-1
Enable zero-crossing detection	on
Initial trigger signal state	compatibility (no trigger on first evaluation)
Port dimensions	-1
Trigger signal sample time	-1
Minimum	[]
Maximum	[]
Data type	Inherit: auto
Interpolate data	on
FunctionPrototype	f()

OBSERVATION

Figure 3.7. warning_system/OBSERVATION



Blocks

Parameters

"acc_calibrated" (From)

Table 3.46. "acc_calibrated" Parameters

Parameter	Value
Goto tag	acc_calibrated
Icon display	Tag

"acc_offset" (From)

Table 3.47. "acc_offset" Parameters

Parameter	Value
Goto tag	acc_offset
Icon display	Tag

"Bus Selector" (BusSelector)

Table 3.48. "Bus Selector" Parameters

Parameter	Value
Output signals	local_x,local_y,local_z
Output as virtual bus	off
InputSignals	local_x local_y local_z

Output Hierarchy:

1. *Bus Selector*
    1. <local_x>
    2. <local_y>
    3. <local_z>

"Bus Selector1" (BusSelector)

Table 3.49. "Bus Selector1" Parameters

Parameter	Value
Output signals	Lat,Lng,Speed
Output as virtual bus	off
InputSignals	Lat Lng Speed

Output Hierarchy:

1. *Bus Selector1*
    1. <Lat>
    2. <Lng>
    3. <Speed>

"Bus Selector10" (BusSelector)

Table 3.50. "Bus Selector10" Parameters

Parameter	Value
Output signals	X,Y,Z
Output as virtual bus	off
InputSignals	X Y Z

Output Hierarchy:

1. *Bus Selector10*
    1. <X>
    2. <Y>
    3. <Z>

"Bus Selector2" (BusSelector)

Table 3.51. "Bus Selector2" Parameters

Parameter	Value
Output signals	X,Y,Z
Output as virtual bus	off
InputSignals	X Y Z

Output Hierarchy:

1. *Bus Selector2*
    1. <X>
    2. <Y>
    3. <Z>

"Bus Selector5" (BusSelector)

Table 3.52. "Bus Selector5" Parameters

Parameter	Value
Output signals	lat,lng,speed,status,address
Output as virtual bus	off

Parameter	Value
InputSignals	lat lng speed address status

Output Hierarchy:

1. *Bus Selector5*
    1. <lat>
    2. <lng>
    3. <speed>
    4. <status>
    5. <address>

"Bus Selector6" (BusSelector)

Table 3.53. "Bus Selector6" Parameters

Parameter	Value
Output signals	gyro_X,gyro_Y,gyro_Z
Output as virtual bus	off
InputSignals	gyro_X gyro_Y gyro_Z

Output Hierarchy:

1. *Bus Selector6*
    1. <gyro_X>
    2. <gyro_Y>
    3. <gyro_Z>

"Bus Selector7" (BusSelector)

Table 3.54. "Bus Selector7" Parameters

Parameter	Value
Output signals	X,Y,Z
Output as virtual bus	off
InputSignals	X Y Z

Output Hierarchy:

1. *Bus Selector7*
    1. <X>
    2. <Y>
    3. <Z>

"Bus Selector8" (BusSelector)

Table 3.55. "Bus Selector8" Parameters

Parameter	Value
Output signals	local_x,local_y,local_z
Output as virtual bus	off
InputSignals	local_x local_y local_z

Output Hierarchy:

1. *Bus Selector8*
    1. <local_x>
    2. <local_y>
    3. <local_z>

"Bus Selector9" (BusSelector)

Table 3.56. "Bus Selector9" Parameters

Parameter	Value
Output signals	X,Y,Z
Output as virtual bus	off
InputSignals	X Y Z

Output Hierarchy:

1. *Bus Selector9*
    1. <X>
    2. <Y>
    3. <Z>

"delta" (From)

Table 3.57. "delta" Parameters

Parameter	Value
Goto tag	DELTA
Icon display	Tag

"delta1" (From)

Table 3.58. "delta1" Parameters

Parameter	Value
Goto tag	DELTA
Icon display	Tag

"Display1" (Display)

Table 3.59. "Display1" Parameters

Parameter	Value
Numeric display format	short
Decimation	1
Floating display	off

"Display2" (Display)

Table 3.60. "Display2" Parameters

Parameter	Value
Numeric display format	short
Decimation	1
Floating display	off

"Display22" (Display)

Table 3.61. "Display22" Parameters

Parameter	Value
Numeric display format	short

Parameter	Value
Decimation	1
Floating display	off

"Display23" (Display)

Table 3.62. "Display23" Parameters

Parameter	Value
Numeric display format	short
Decimation	1
Floating display	off

"Display24" (Display)

Table 3.63. "Display24" Parameters

Parameter	Value
Numeric display format	short
Decimation	1
Floating display	off

"Display3" (Display)

Table 3.64. "Display3" Parameters

Parameter	Value
Numeric display format	short
Decimation	1
Floating display	off

"Display4" (Display)

Table 3.65. "Display4" Parameters

Parameter	Value
Numeric display format	short
Decimation	1
Floating display	off

"Display5" (Display)

Table 3.66. "Display5" Parameters

Parameter	Value
Numeric display format	short
Decimation	1
Floating display	off

"From1" (From)

Table 3.67. "From1" Parameters

Parameter	Value
Goto tag	local_gyro
Icon display	Tag

"From3" (From)

Table 3.68. "From3" Parameters

Parameter	Value
Goto tag	local_gps
Icon display	Tag

"From4" (From)

Table 3.69. "From4" Parameters

Parameter	Value
Goto tag	local_acc
Icon display	Tag

"From5" (From)

Table 3.70. "From5" Parameters

Parameter	Value
Goto tag	FOREIGN_GPS
Icon display	Tag

"local_acc" (From)

Table 3.71. "local_acc" Parameters

Parameter	Value
Goto tag	local_acc
Icon display	Tag

"Mux" (Mux)

Table 3.72. "Mux" Parameters

Parameter	Value
Number of inputs	2
Display option	bar

"Mux1" (Mux)

Table 3.73. "Mux1" Parameters

Parameter	Value
Number of inputs	2
Display option	bar

"Mux2" (Mux)

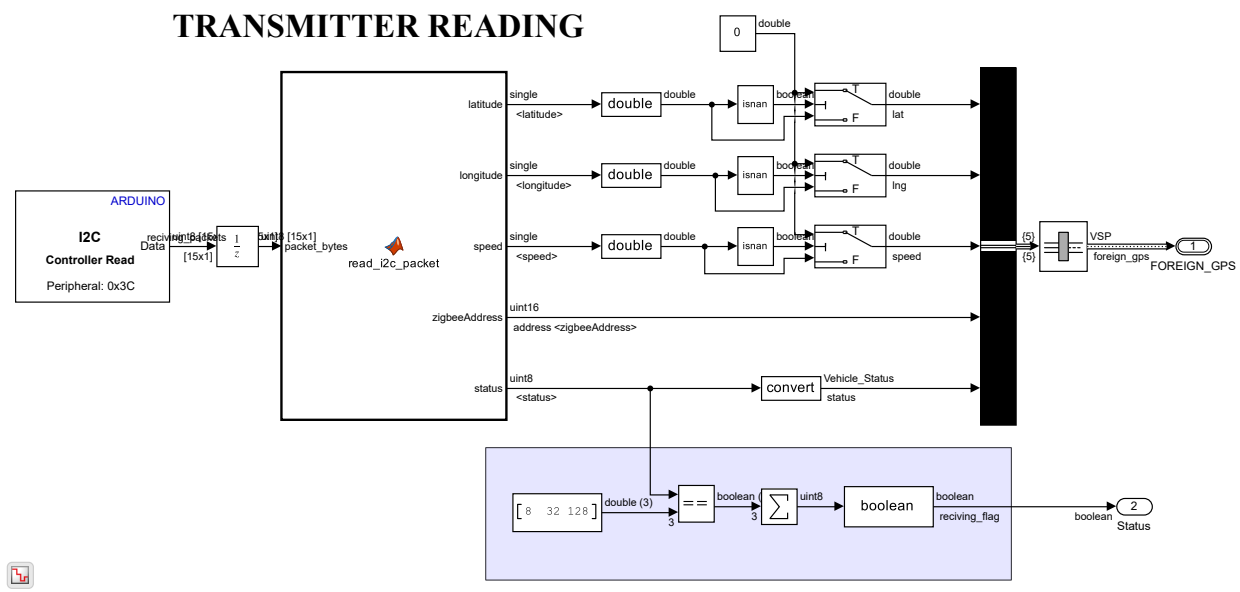
Table 3.74. "Mux2" Parameters

Parameter	Value
Number of inputs	2
Display option	bar

TRANSMITTER_READING

Checksum: 2132600380 489520529 1035083572 1203551146

Figure 3.8. warning_system/TRANSMITTER_READING



Interface

Output Signals

The following tables describe the signals output by this system. The name of the output signal is the name of the signal's parent block, i.e., the block that computes the signal. The number in angle brackets is the number of the port that emits the signal.

Table 3.75. Output Signals

Signal Name	Block	Description	Data Type	Width	Dimensions
foreign_gps	warning_system/TRANSMITTER_READING/Signal Conversion		VSP	1	1x1
reciving_flag	warning_system/TRANSMITTER_READING/Cast To Boolean		boolean	1	1x1

Blocks

Parameters

"Bus Creator" (BusCreator)

Table 3.76. "Bus Creator" Parameters

Parameter	Value
Number of inputs	5
Display option	bar
Data type	Bus: VSP
Output as nonvirtual bus	off
Use names from inputs instead of from bus object	on

"Cast To Boolean" (DataTypeConversion)

Table 3.77. "Cast To Boolean" Parameters

Parameter	Value
Output minimum	[]
Output maximum	[]
Output data type	boolean
Lock output data type setting against changes by the fixed-point tools	off
Input and output to have equal	Real World Value (RWV)
Integer rounding mode	Floor
Saturate on integer overflow	off
Sample time (-1 for inherited)	-1

"Constant" (Constant)

Table 3.78. "Constant" Parameters

Parameter	Value
Constant value	0
Interpret vector parameters as 1-D	on
Output minimum	[]
Output maximum	[]
Output data type	double
Lock output data type setting against changes by the fixed-point tools	off
Sample time	inf
Frame period	inf

"Constant1" (Constant)

Table 3.79. "Constant1" Parameters

Parameter	Value
Constant value	[8 32 128]
Interpret vector parameters as 1-D	on
Output minimum	[]
Output maximum	[]
Output data type	Inherit: Inherit from 'Constant value'
Lock output data type setting against changes by the fixed-point tools	off
Sample time	inf
Frame period	inf

"Data Type Conversion1" (DataTypeConversion)

Table 3.80. "Data Type Conversion1" Parameters

Parameter	Value
Output minimum	[]
Output maximum	[]
Output data type	Enum: Vehicle_Status
Lock output data type setting against changes by the fixed-point tools	off
Input and output to have equal	Real World Value (RWV)
Integer rounding mode	Floor
Saturate on integer overflow	off
Sample time (-1 for inherited)	-1

"Data Type Conversion2" (DataTypeConversion)

Table 3.81. "Data Type Conversion2" Parameters

Parameter	Value
Output minimum	[]
Output maximum	[]
Output data type	double
Lock output data type setting against changes by the fixed-point tools	off
Input and output to have equal	Real World Value (RWV)

Parameter	Value
Integer rounding mode	Floor
Saturate on integer overflow	off
Sample time (-1 for inherited)	-1

"Data Type Conversion3" (DataTypeConversion)

Table 3.82. "Data Type Conversion3" Parameters

Parameter	Value
Output minimum	[]
Output maximum	[]
Output data type	double
Lock output data type setting against changes by the fixed-point tools	off
Input and output to have equal	Real World Value (RWV)
Integer rounding mode	Floor
Saturate on integer overflow	off
Sample time (-1 for inherited)	-1

"Data Type Conversion4" (DataTypeConversion)

Table 3.83. "Data Type Conversion4" Parameters

Parameter	Value
Output minimum	[]
Output maximum	[]
Output data type	double
Lock output data type setting against changes by the fixed-point tools	off
Input and output to have equal	Real World Value (RWV)
Integer rounding mode	Floor
Saturate on integer overflow	off
Sample time (-1 for inherited)	-1

"FOREIGN_GPS" (Outport)

Table 3.84. "FOREIGN_GPS" Parameters

Parameter	Value
Port number	1

Parameter	Value
Icon display	Port number
Output function call	off
Minimum	[]
Maximum	[]
Data type	Bus: VSP
Lock output data type setting against changes by the fixed-point tools	off
Output as nonvirtual bus in parent model	off
Bus virtuality	inherit
Data mode	inherit
Unit (e.g., m, m/s^2, N*m)	inherit
Port dimensions (-1 for inherited)	-1
Variable-size signal	Inherit
Sample time (-1 for inherited)	-1
Ensure outport is virtual	off
Output when disabled	held
Initial output	[]
MustResolveToSignalObject	off
Specify output when source is unconnected	off
Constant value	0
Interpret vector parameters as 1-D	on

"I2C Read" (MATLABSystem)

Table 3.85. "I2C Read" Parameters

Parameter	Value
arduinoblockmaskI2CModuleProp	0
arduinoblockmaskSlaveAddressProp	hex2dec('3C')
arduinoblockmaskSlaveByteOrderProp	BigEndian
arduinoblockmaskRegisterAddressModeProp	off
arduinoblockmaskSlaveDataTypeProp	uint8
arduinoblockmaskDataLengthProp	15
arduinoblockmaskOutputStatusProp	off
arduinoblockmaskSampleTimeProp	TS_READER

"IsNaN" (RelationalOperator)

Table 3.86. "IsNaN" Parameters

Parameter	Value
Relational operator	isNaN
Require all inputs to have the same data type	off
Output data type	boolean
Enable zero-crossing detection	on
Sample time (-1 for inherited)	-1
Integer rounding mode	Simplest

"IsNaN1" (RelationalOperator)

Table 3.87. "IsNaN1" Parameters

Parameter	Value
Relational operator	isNaN
Require all inputs to have the same data type	off
Output data type	boolean
Enable zero-crossing detection	on
Sample time (-1 for inherited)	-1
Integer rounding mode	Simplest

"IsNaN2" (RelationalOperator)

Table 3.88. "IsNaN2" Parameters

Parameter	Value
Relational operator	isNaN
Require all inputs to have the same data type	off
Output data type	boolean
Enable zero-crossing detection	on
Sample time (-1 for inherited)	-1
Integer rounding mode	Simplest

"MATLAB Function" (MATLAB Function)

Table 3.89. MATLAB Function Function Properties

Property	Value
Update Method	INHERITED
Sample Time	-1
Support variable-size arrays	1
Saturate on integer overflow	1
Treat these inherited Simulink signal types as fi objects	Fixed-point
MATLAB Function block fimath	Same as MATLAB Default
Input fi math	fimath(...)
Description	

Table 3.90. MATLAB Function Argument Summary

Name	Scope	Port	Data Type	Size
packet_bytes	Input	1	uint8	[15, 1]
latitude	Output	1	single	1
longitude	Output	2	single	1
speed	Output	3	single	1
zigbeeAddress	Output	4	uint16	1
status	Output	5	uint8	1

MATLAB Function Function Script

```
function [latitude, longitude, speed, zigbeeAddress, status] = read_i2c_pack-
et(packet_bytes)
    %#codegen

    % Define output sizes and types
    latitude = single(0);
    longitude = single(0);
    speed = single(0);
    status = uint8(0);
    zigbeeAddress = uint16(0);

    % Validate packet size
    if length(packet_bytes) ~= 15
```

```
        return; % Return default values if packet size is incorrect
    end

    % Extract byte segments and convert back to original data types
    latitude = typecast(uint8(packet_bytes(1:4)), 'single');
    longitude = typecast(uint8(packet_bytes(5:8)), 'single');
    speed = typecast(uint8(packet_bytes(9:12)), 'single');
    zigbeeAddress = typecast(uint8(packet_bytes(13:14)), 'uint16'); % Recon-
struct as uint16
    status = uint8(packet_bytes(15)); % Already uint8, no need for typecast
end
```

Table 3.91. MATLAB Function Supporting Functions

Function	Defined By	Path
feval	MATLAB	
isstring	MATLAB	
typecast	MATLAB	

"Relational Operator" (RelationalOperator)

Table 3.92. "Relational Operator" Parameters

Parameter	Value
Relational operator	==
Require all inputs to have the same data type	off
Output data type	boolean
Enable zero-crossing detection	on
Sample time (-1 for inherited)	-1
Integer rounding mode	Simplest

"Signal Conversion" (SignalConversion)

Table 3.93. "Signal Conversion" Parameters

Parameter	Value
Output	Nonvirtual bus
Data type	Bus: VSP
Exclude this block from 'Block reduction' optimization	off

"Status" (Outport)

Table 3.94. "Status" Parameters

Parameter	Value
Port number	2
Icon display	Port number
Output function call	off
Minimum	[]
Maximum	[]
Data type	boolean
Lock output data type setting against changes by the fixed-point tools	off
Output as nonvirtual bus in parent model	off
Bus virtuality	inherit
Data mode	inherit
Unit (e.g., m, m/s^2, N*m)	inherit
Port dimensions (-1 for inherited)	-1
Variable-size signal	Inherit
Sample time (-1 for inherited)	-1
Ensure outport is virtual	off
Output when disabled	held
Initial output	[]
MustResolveToSignalObject	off
Specify output when source is unconnected	off
Constant value	0
Interpret vector parameters as 1-D	on

"Sum of Elements" (Sum)

Table 3.95. "Sum of Elements" Parameters

Parameter	Value
Icon shape	rectangular
List of signs	+
Apply over	All dimensions
Dimension	1
Output minimum	[]
Output maximum	[]

Parameter	Value
Output data type	Inherit: Inherit via internal rule
Accumulator data type	Inherit: Inherit via internal rule
Require all inputs to have the same data type	off
Lock data type settings against changes by the fixed-point tools	off
Integer rounding mode	Floor
Saturate on integer overflow	off
Sample time (-1 for inherited)	-1

"Switch" (Switch)

Table 3.96. "Switch" Parameters

Parameter	Value
Criteria for passing first input	u2 ~= 0
Threshold	0
Require all data port inputs to have the same data type	off
Output minimum	[]
Output maximum	[]
Output data type	Inherit: Inherit via intern al rule
Lock output data type setting against changes by the fixed-point tools	off
Integer rounding mode	Floor
Saturate on integer overflow	off
Enable zero-crossing detection	on
Sample time (-1 for inherited)	-1
Allow different data input sizes (Results in variable-size output signal)	off

"Switch1" (Switch)

Table 3.97. "Switch1" Parameters

Parameter	Value
Criteria for passing first input	u2 ~= 0
Threshold	0
Require all data port inputs to have the same data type	off
Output minimum	[]
Output maximum	[]

Parameter	Value
Output data type	Inherit: Inherit via intern al rule
Lock output data type setting against changes by the fixed-point tools	off
Integer rounding mode	Floor
Saturate on integer overflow	off
Enable zero-crossing detection	on
Sample time (-1 for inherited)	-1
Allow different data input sizes (Results in variable-size output signal)	off

"Switch2" (Switch)

Table 3.98. "Switch2" Parameters

Parameter	Value
Criteria for passing first input	u2 ~= 0
Threshold	0
Require all data port inputs to have the same data type	off
Output minimum	[]
Output maximum	[]
Output data type	Inherit: Inherit via intern al rule
Lock output data type setting against changes by the fixed-point tools	off
Integer rounding mode	Floor
Saturate on integer overflow	off
Enable zero-crossing detection	on
Sample time (-1 for inherited)	-1
Allow different data input sizes (Results in variable-size output signal)	off

"Unit Delay" (UnitDelay)

Table 3.99. "Unit Delay" Parameters

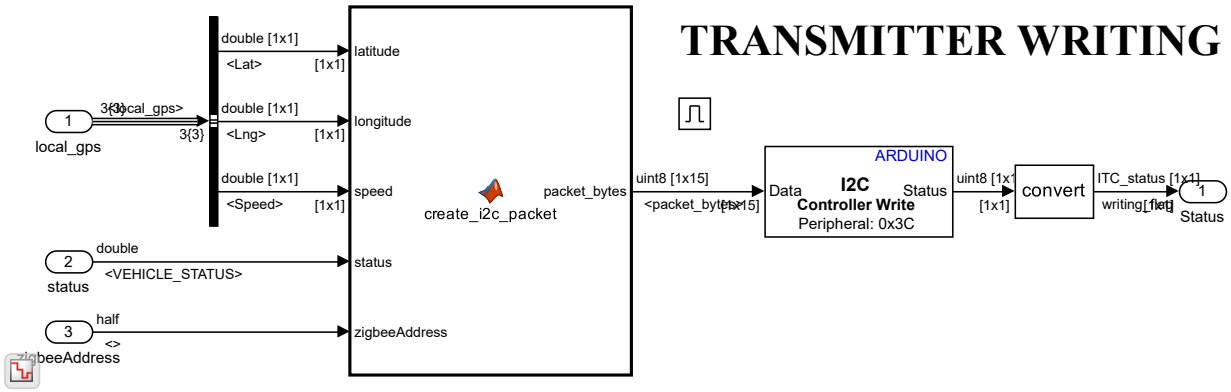
Parameter	Value
Initial condition	0
Input processing	Elements as channels (sample based)
Sample time (-1 for inherited)	-1
State name must resolve to Simulink signal object	off

Block Execution Order

1. Constant (Constant)
2. Constant1 (Constant)
3. Unit Delay (UnitDelay)
4. *MATLAB Function*
   1. SFunction (S-Function)
5. Relational Operator (RelationalOperator)
6. Sum of Elements (Sum)
7. Cast To Boolean (DataTypeConversion)
8. Data Type Conversion1 (DataTypeConversion)
9. Data Type Conversion2 (DataTypeConversion)
10. Data Type Conversion3 (DataTypeConversion)
11. Data Type Conversion4 (DataTypeConversion)
12. I2C Read (MATLABSystem)
13. IsNaN (RelationalOperator)
14. IsNaN1 (RelationalOperator)
15. IsNaN2 (RelationalOperator)
16. Switch (Switch)
17. Switch1 (Switch)
18. Switch2 (Switch)
19. Signal Conversion_BusCreator1 (BusCreator)

TRANSMITTER_WRITTING

Figure 3.9. warning_system/TRANSMITTER_WRITTING



Blocks

Parameters

"Bus Selector2" (BusSelector)

Table 3.100. "Bus Selector2" Parameters

Parameter	Value
Output signals	Lat,Lng,Speed
Output as virtual bus	off
InputSignals	Lat Lng Speed

Output Hierarchy:

1. *Bus Selector2*
    1. <Lat>
    2. <Lng>
    3. <Speed>

"Data Type Conversion" (DataTypeConversion)

Table 3.101. "Data Type Conversion" Parameters

Parameter	Value
Output minimum	[]
Output maximum	[]
Output data type	Enum: ITC_status
Lock output data type setting against changes by the fixed-point tools	off
Input and output to have equal	Real World Value (RWV)
Integer rounding mode	Floor
Saturate on integer overflow	off
Sample time (-1 for inherited)	-1

"Enable" (EnablePort)

Table 3.102. "Enable" Parameters

Parameter	Value
States when enabling	held
Propagate sizes of variable-size signals	Only when enabling
Show output port	off
Enable zero-crossing detection	on

Parameter	Value
Port dimensions	1
Sample time	-1
Minimum	[]
Maximum	[]
Data type	double
Interpolate data	on

"I2C Write" (MATLABSystem)

Table 3.103. "I2C Write" Parameters

Parameter	Value
arduinoblockmaskI2CModuleProp	0
arduinoblockmaskSlaveAddressProp	hex2dec('3C')
arduinoblockmaskSlaveByteOrderProp	BigEndian
arduinoblockmaskRegisterAddressModeProp	off
arduinoblockmaskRepeatedStartProp	off
arduinoblockmaskOutputStatusProp	on

"local_gps" (Inport)

Table 3.104. "local_gps" Parameters

Parameter	Value
Port number	1
Port dimensions (-1 for inherited)	-1
Sample time (-1 for inherited)	-1
Minimum	[]
Maximum	[]
Data type	Inherit: auto

"MATLAB Function" (MATLAB Function)

Table 3.105. MATLAB Function Function Properties

Property	Value
Update Method	INHERITED
Sample Time	-1

Property	Value
Support variable-size arrays	1
Saturate on integer overflow	1
Treat these inherited Simulink signal types as fi objects	Fixed-point
MATLAB Function block fimath	Same as MATLAB Default
Input fi math	fimath(...)
Description	

Table 3.106. MATLAB Function Argument Summary

Name	Scope	Port	Data Type	Size
latitude	Input	1	double	1
packet_bytes	Output	1	uint8	[1, 15]
longitude	Input	2	double	1
speed	Input	3	double	1
status	Input	4	double	1
zigbeeAddress	Input	5	real16_T	1

MATLAB Function Function Script

```
function packet_bytes = create_i2c_packet(latitude, longitude, speed, sta-
tus, zigbeeAddress)
    %#codegen

    % Define fixed-size output (1x15 uint8 array)
    packet_bytes = zeros(1, 15, 'uint8');

    % single --> float32
    % double --> float64

    % Convert to byte representation
    lat_bytes = typecast(single(latitude), 'uint8'); % 4 byte
    lon_bytes = typecast(single(longitude), 'uint8'); % 4 byte
    speed_bytes = typecast(single(speed), 'uint8'); % 4 byte
    status_byte = uint8(status); % 1 byte
    zigbee_bytes = typecast(uint16(zigbeeAddress), 'uint8'); % 2 bytes

    % Store in the output array
    packet_bytes(1:4)   = lat_bytes;
    packet_bytes(5:8)   = lon_bytes;
```

```
    packet_bytes(9:12)  = speed_bytes;
    packet_bytes(13:14) = zigbee_bytes;
    packet_bytes(15)    = status_byte; % Adjusted for uint32
end
```

Table 3.107. MATLAB Function Supporting Functions

Function	Defined By	Path
feval	MATLAB	
intmax	MATLAB	
typecast	MATLAB	

"Status" (Outport)

Table 3.108. "Status" Parameters

Parameter	Value
Port number	1
Icon display	Port number
Output function call	off
Minimum	[]
Maximum	[]
Data type	Inherit: auto
Lock output data type setting against changes by the fixed-point tools	off
Output as nonvirtual bus in parent model	off
Bus virtuality	inherit
Data mode	inherit
Unit (e.g., m, m/s^2, N*m)	inherit
Port dimensions (-1 for inherited)	-1
Variable-size signal	Inherit
Sample time (-1 for inherited)	-1
Ensure outport is virtual	off
Output when disabled	held
Initial output	[]
MustResolveToSignalObject	off
Specify output when source is unconnected	off
Constant value	0
Interpret vector parameters as 1-D	on

"status" (Inport)

Table 3.109. "status" Parameters

Parameter	Value
Port number	2
Port dimensions (-1 for inherited)	-1
Sample time (-1 for inherited)	-1
Minimum	[]
Maximum	[]
Data type	Inherit: auto

"zigbeeAddress" (Inport)

Table 3.110. "zigbeeAddress" Parameters

Parameter	Value
Port number	3
Port dimensions (-1 for inherited)	-1
Sample time (-1 for inherited)	-1
Minimum	[]
Maximum	[]
Data type	Inherit: auto

Chapter 4. System Design Variables

Design Variable Summary

Table 4.1. Design Variables

Variable Name	Parent Blocks	Size	Bytes	Class	Value
ACCEL_RANGE	MPU6050	1x1	1	uint8	4
DLPF_SETTING	MPU6050	1x1	1	uint8	6
ENABLE_GYRO	MPU6050	1x1	1	boolean	1
GYRO_RANGE	MPU6050	1x1	2	uint16	250
THRESHOLD_CAUTION	CORE_LOGIC	1x1	8	double	0.6000
THRESHOLD_CRASHED	CORE_LOGIC	1x1	8	double	0.8000
THRESHOLD_LANE_CHANGE	CORE_LOGIC	1x1	8	double	0.2000
THRESHOLD_REST	CORE_LOGIC	1x1	8	double	0.2000
TS_CORE	CORE_LOGIC	1x1	8	double	0.0500
TS_GPS	GPS_PARSER	1x1	8	double	0.0500
TS_MPU	MPU6050	1x1	8	double	0.0500
TS_PRINT	MATLAB Function	1x1	8	double	0.2000
TS_READER	I2C Read	1x1	8	double	0.2000
TS_WRITTER	MATLAB Function	1x1	8	double	0.2000
VSP	MATLAB Function Bus Creator FOREIGN_GPS Signal Conversion	1x1	333	Simulink.Bus	<Simulink.Bus>

Variable Name	Parent Blocks	Size	Bytes	Class	Value
accel	CORE_LOGIC Bus Creator Signal Conversion offset raw_acc Bus Creator Signal Conversion1 calibrated_acc delta Signal Conversion Signal Conversion1	1x1	167	Simulink.Bus	<Simulink.Bus>

Table 4.2. Functions used in Design Variable Expressions

Function Name	Parent Blocks	Calling character vector
hex2dec	I2C Read I2C Read Ecriture I2C I2C Read I2C Write	hex2dec('3b') hex2dec('68') hex2dec('68') hex2dec('3C') hex2dec('3C')

Design Variable Details

Table 4.3. ACCEL_RANGE

Property	Value
Value	4
Complexity	real
Dimensions	[1 1]
CoderInfo	ACCEL_RANGE.CoderInfo
Description	
DataType	uint8
Min	
Max	
Unit	

Table 4.4. ACCEL_RANGE.CoderInfo

Property	Value

StorageClass	ExportedGlobal
TypeQualifier	
Identifier	
Alignment	-1
CustomStorageClass	Default
CustomAttributes	ACCEL_RANGE.CoderInfo.CustomAttributes

ACCEL_RANGE.CoderInfo.CustomAttributes (SimulinkCSC.AttribClass_Simulink_Default,)
Note: this object has no unfiltered properties.

Used by Blocks:

- warning_system/MPU6050

Resolved in: data dictionary (try.sldd)

Table 4.5. DLPF_SETTING

Property	Value
Value	6
Complexity	real
Dimensions	[1 1]
CoderInfo	DLPF_SETTING.CoderInfo
Description	
DataType	uint8
Min	
Max	
Unit	

Table 4.6. DLPF_SETTING.CoderInfo

Property	Value
StorageClass	ExportedGlobal
TypeQualifier	
Identifier	
Alignment	-1
CustomStorageClass	Default
CustomAttributes	DLPF_SETTING.CoderInfo.CustomAttributes

DLPF_SETTING.CoderInfo.CustomAttributes (SimulinkCSC.AttribClass_Simulink_Default,)

Note: this object has no unfiltered properties.

Used by Blocks:

- warning_system/MPU6050

Resolved in: data dictionary (try.sldd)

Table 4.7. ENABLE_GYRO

Property	Value
Value	true
Complexity	real
Dimensions	[1 1]
CoderInfo	ENABLE_GYRO.CoderInfo
Description	
DataType	boolean
Min	
Max	
Unit	

Table 4.8. ENABLE_GYRO.CoderInfo

Property	Value
StorageClass	ExportedGlobal
TypeQualifier	
Identifier	
Alignment	-1
CustomStorageClass	Default
CustomAttributes	ENABLE_GYRO.CoderInfo.CustomAttributes

ENABLE_GYRO.CoderInfo.CustomAttributes (SimulinkCSC.AttribClass_Simulink_Default,)

Note: this object has no unfiltered properties.

Used by Blocks:

- warning_system/MPU6050

Resolved in: data dictionary (try.sldd)

Table 4.9. GYRO_RANGE

Property	Value
Value	250

Complexity	real
Dimensions	[1 1]
CoderInfo	GYRO_RANGE.CoderInfo
Description	
DataType	uint16
Min	
Max	
Unit	

Table 4.10. GYRO_RANGE.CoderInfo

Property	Value
StorageClass	ExportedGlobal
TypeQualifier	
Identifier	
Alignment	-1
CustomStorageClass	Default
CustomAttributes	GYRO_RANGE.CoderInfo.CustomAttributes

GYRO_RANGE.CoderInfo.CustomAttributes (SimulinkCSC.AttribClass_Simulink_Default,)
Note: this object has no unfiltered properties.

Used by Blocks:

- warning_system/MPU6050

Resolved in: data dictionary (try.sldd)

THRESHOLD_CAUTION. 0.6000

Used by Blocks:

- warning_system/CORE_LOGIC

Resolved in: data dictionary (try.sldd)

THRESHOLD_CRASHED. 0.8000

Used by Blocks:

- warning_system/CORE_LOGIC

Resolved in: data dictionary (try.sldd)

THRESHOLD_LANE_CHANGE. 0.2000

Used by Blocks:

- warning_system/CORE_LOGIC

Resolved in: data dictionary (try.sldd)

THRESHOLD_REST. 0.2000

Used by Blocks:

- warning_system/CORE_LOGIC

Resolved in: data dictionary (try.sldd)

TS_CORE. 0.0500

Used by Blocks:

- warning_system/CORE_LOGIC

Resolved in: data dictionary (try.sldd)

TS_GPS. 0.0500

Used by Blocks:

- warning_system/GPS_PARSER

Resolved in: data dictionary (try.sldd)

TS_MPU. 0.0500

Used by Blocks:

- warning_system/MPU6050

Resolved in: data dictionary (try.sldd)

TS_PRINT. 0.2000

Used by Blocks:

- warning_system/ serial_printing/MATLAB Function

Resolved in: data dictionary (try.sldd)

TS_READER. 0.2000

Used by Blocks:

- warning_system/TRANSMITTER_READING/I2C Read

Resolved in: data dictionary (try.sldd)

TS_WRITTER. 0.2000

Used by Blocks:

- warning_system/TRANSMITTER_WRITTING/MATLAB Function

Resolved in: data dictionary (try.sldd)

Table 4.11. VSP

Property	Value
Alignment	-1
PreserveElementDimensions	false
Elements	[VSP.Elements(1), VSP.Elements(2), VSP.Elements(3), VSP.Elements(4), VSP.Elements(5)]
Description	
DataScope	Auto
HeaderFile	

Table 4.12. VSP.Elements(1)

Property	Value
Min	
Max	
DimensionsMode	Fixed
Description	
Unit	
Name	lat
DataType	double
Complexity	real
Dimensions	1

Table 4.13. VSP.Elements(2)

Property	Value
Min	
Max	
DimensionsMode	Fixed
Description	
Unit	
Name	lng
DataType	double
Complexity	real
Dimensions	1

Table 4.14. VSP.Elements(3)

Property	Value
Min	
Max	
DimensionsMode	Fixed
Description	
Unit	
Name	speed
DataType	double
Complexity	real
Dimensions	1

Table 4.15. VSP.Elements(4)

Property	Value
Min	
Max	
DimensionsMode	Fixed
Description	
Unit	
Name	address
DataType	uint16
Complexity	real
Dimensions	1

Table 4.16. VSP.Elements(5)

Property	Value
Min	
Max	
DimensionsMode	Fixed
Description	
Unit	
Name	status
DataType	Enum: Vehicle_Status
Complexity	real

Dimensions	1

Used by Blocks:

- warning_system/ serial_printing/MATLAB Function
- warning_system/TRANSMITTER_READING/Bus Creator
- warning_system/TRANSMITTER_READING/FOREIGN_GPS
- warning_system/TRANSMITTER_READING/Signal Conversion

Resolved in: data dictionary (try.sldd)

Table 4.17. accel

Property	Value
Alignment	-1
PreserveElementDimensions	false
Elements	[accel.Elements(1), accel.Elements(2), accel.Elements(3)]
Description	
DataScope	Auto
HeaderFile	

Table 4.18. accel.Elements(1)

Property	Value
Min	
Max	
DimensionsMode	Fixed
Description	
Unit	
Name	X
DataType	double
Complexity	real
Dimensions	1

Table 4.19. accel.Elements(2)

Property	Value
Min	
Max	
DimensionsMode	Fixed
Description	

Unit	
Name	Y
DataType	double
Complexity	real
Dimensions	1

Table 4.20. accel.Elements(3)

Property	Value
Min	
Max	
DimensionsMode	Fixed
Description	
Unit	
Name	Z
DataType	double
Complexity	real
Dimensions	1

Used by Blocks:

- warning_system/CORE_LOGIC
- warning_system/CORE_LOGIC/calibrate/Bus Creator
- warning_system/CORE_LOGIC/calibrate/Signal Conversion
- warning_system/CORE_LOGIC/calibrate/offset
- warning_system/CORE_LOGIC/calibrate/raw_acc
- warning_system/CORE_LOGIC/check_jerk/Bus Creator
- warning_system/CORE_LOGIC/check_jerk/Signal Conversion1
- warning_system/CORE_LOGIC/check_jerk/calibrated_acc
- warning_system/CORE_LOGIC/check_jerk/delta
- warning_system/Signal Conversion
- warning_system/Signal Conversion1

Resolved in: data dictionary (try.sldd)

Enumeration Types

Table 4.21. ITC_status Properties

Property	Value
Name	ITC_status

Property	Value
Source Type	data dictionary
Source	try.sldd
Members	• SUCCESS (0) • OCCUPIED (1) • ARBITRATION_LOST (2) • NO_ACKNOWLEDGE (4) • BUS_ERROR (8) • BUS_NOT_IN_USE (16)
Default Value	NO_ACKNOWLEDGE (4)

Table 4.22. Vehicle_Status Properties

Property	Value
Name	Vehicle_Status
Source Type	data dictionary
Source	try.sldd
Members	• INVALID (0) • NORMAL (4) • LANE_CHANGE_RIGHT (8) • LANE_CHANGE_LEFT (16) • NEARBY_CRASH (32) • CAUTION (100) • CRASH (128)
Default Value	INVALID (0)

Chapter 5. System Model Configuration

Source: Model
Source Name: warning_system

Table 5.1. warning_system Configuration Set

Property	Value
Description	
Components	[warning_system Configuration Set.Components(1), warning_system Configuration Set.Components(2), warning_system Configuration Set.Components(3), warning_system Configuration Set.Components(4), warning_system Configuration Set.Components(5), warning_system Configuration Set.Components(6), warning_system Configuration Set.Components(7), warning_system Configuration Set.Components(8), warning_system Configuration Set.Components(9), warning_system Configuration Set.Components(10)]
Name	Configuration

Table 5.2. warning_system Configuration Set.Components(1)

Property	Value
Name	Solver
Description	
Components	
StartTime	0.0
StopTime	inf
AbsTol	auto
AutoScaleAbsTol	on
FixedStep	TS_MODEL
InitialStep	auto
MaxOrder	5
ZcThreshold	auto
ConsecutiveZCsStepRelTol	10*128*eps
MaxConsecutiveZCs	1000
ExtrapolationOrder	4
NumberNewtonIterations	1

MaxStep	auto
MinStep	auto
MaxConsecutiveMinStep	1
RelTol	1e-3
EnableMultiTasking	off
AllowMultiTaskInputOutput	off
ConcurrentTasks	off
SolverName	FixedStepAuto
SolverType	Fixed-step
SolverJacobianMethodControl	auto
DaesscMode	auto
ShapePreserveControl	DisableAll
ZeroCrossControl	UseLocalSettings
ZeroCrossAlgorithm	Nonadaptive
SolverResetMethod	Fast
PositivePriorityOrder	off
AutoInsertRateTranBlk	on
SampleTimeConstraint	Unconstrained
InsertRTBMode	Whenever possible
SampleTimeProperty	
DecoupledContinuousIntegration	off
MinimalZcImpactIntegration	off
ODENIntegrationMethod	ode3
EnableFixedStepZeroCrossing	off
MaxZcPerStep	2
MaxZcBracketingIterations	10

Table 5.3. warning_system Configuration Set.Components(2)

Property	**Value**
Name	Data Import/Export
Description	
Components	
Decimation	1
ExternalInput	[t, u]
FinalStateName	xFinal
InitialState	xInitial

LimitDataPoints	off
MaxDataPoints	1000
LoadExternalInput	off
LoadInitialState	off
SaveFinalState	off
SaveOperatingPoint	off
SaveFormat	Dataset
SaveOutput	on
SaveState	off
SignalLogging	on
DSMLogging	on
StreamToWks	on
InspectSignalLogs	off
SaveTime	on
ReturnWorkspaceOutputs	on
StateSaveName	xout
TimeSaveName	tout
OutputSaveName	yout
SignalLoggingName	logsout
DSMLoggingName	dsmout
OutputOption	RefineOutputTimes
OutputTimes	[]
ReturnWorkspaceOutputsName	out
Refine	1
LoggingToFile	off
DatasetSignalFormat	timeseries
LoggingFileName	out.mat
LoggingIntervals	[-inf, inf]

Table 5.4. warning_system Configuration Set.Components(3)

Property	**Value**
Name	Optimization
Description	
Components	
BlockReduction	on
BooleanDataType	on

ConditionallyExecuteInputs	on
DefaultParameterBehavior	Tunable
InlineParams	off
UseDivisionForNetSlopeComputation	off
GainParamInheritBuiltInType	off
UseFloatMulNetSlope	off
InheritOutputTypeSmallerThanSingle	off
DefaultUnderspecifiedDataType	double
UseSpecifiedMinMax	off
InlineInvariantSignals	off
OptimizeBlockIOStorage	on
BufferReuse	on
ReuseModelBlockBuffer	on
GlobalBufferReuse	on
GlobalVariableUsage	Use global to hold temporary results
StrengthReduction	off
AdvancedOptControl	
ExpressionFolding	on
BooleansAsBitfields	off
BitfieldContainerType	uint_T
BitwiseOrLogicalOp	Same as modeled
EnableMemcpy	on
MemcpyThreshold	64
PassReuseOutputArgsAs	Individual arguments
PassReuseOutputArgsThreshold	12
LocalBlockOutputs	on
RollThreshold	5
StateBitsets	off
DataBitsets	off
ActiveStateOutputEnumStorageType	Native Integer
ZeroExternalMemoryAtStartup	off
ZeroInternalMemoryAtStartup	off
InitFltsAndDblsToZero	off
NoFixptDivByZeroProtection	off
EfficientFloat2IntCast	off
EfficientMapNaN2IntZero	on

LifeSpan	auto
EvaledLifeSpan	1
ClockResolution	-1
MaxStackSize	64
BufferReusableBoundary	on
RemoveLocalVariableInitialization	on
SimCompilerOptimization	off
AccelVerboseBuild	off
OptimizeBlockOrder	speed
OptimizeDataStoreBuffers	on
BusAssignmentInplaceUpdate	on
DifferentSizesBufferReuse	on
UseRowMajorAlgorithm	off
OptimizationLevel	level2
OptimizationPriority	Balanced
OptimizationCustomize	off
LabelGuidedReuse	off
MultiThreadedLoops	off
AutoScheduleForLoops	off
DenormalBehavior	GradualUnderflow
EfficientTunableParamExpr	on

Table 5.5. warning_system Configuration Set.Components(4)

Property	**Value**
Name	Diagnostics
Description	
Components	
RTPrefix	error
ConsistencyChecking	none
ArrayBoundsChecking	none
SignalInfNanChecking	none
StringTruncationChecking	error
SignalRangeChecking	none
ReadBeforeWriteMsg	UseLocalSettings
WriteAfterWriteMsg	UseLocalSettings
WriteAfterReadMsg	UseLocalSettings

AlgebraicLoopMsg	warning
ArtificialAlgebraicLoopMsg	warning
SaveWithDisabledLinksMsg	warning
SaveWithParameterizedLinksMsg	warning
CheckSSInitialOutputMsg	on
UnderspecifiedInitializationDetection	Simplified
MergeDetectMultiDrivingBlocksExec	error
SignalResolutionControl	UseLocalSettings
BlockPriorityViolationMsg	warning
MinStepSizeMsg	warning
TimeAdjustmentMsg	none
MaxConsecutiveZCsMsg	error
MaskedZcDiagnostic	warning
IgnoredZcDiagnostic	warning
SolverPrmCheckMsg	none
InheritedTsInSrcMsg	warning
MultiTaskDSMMsg	error
MultiTaskCondExecSysMsg	error
MultiTaskRateTransMsg	error
SingleTaskRateTransMsg	none
TasksWithSamePriorityMsg	warning
SigSpecEnsureSampleTimeMsg	warning
CheckMatrixSingularityMsg	none
IntegerOverflowMsg	warning
Int32ToFloatConvMsg	warning
ParameterDowncastMsg	error
ParameterOverflowMsg	error
ParameterUnderflowMsg	none
ParameterPrecisionLossMsg	warning
ParamSuppressDoubleToSinglePrecisionLossMsg	off
ParamPrecisionLossAbsoluteDiffThreshold	0.0
ParamPrecisionLossRelativeDiffThreshold	0.0
ParamOverflowErrorThreshold	OneBit
ParameterTunabilityLossMsg	error
FixptConstUnderflowMsg	none
FixptConstOverflowMsg	none

FixptConstPrecisionLossMsg	none
UnderSpecifiedDataTypeMsg	none
UnnecessaryDatatypeConvMsg	none
VectorMatrixConversionMsg	none
FcnCallInpInsideContextMsg	error
SignalLabelMismatchMsg	none
UnconnectedInputMsg	none
UnconnectedOutputMsg	none
UnconnectedLineMsg	none
UseOnlyExistingSharedCode	error
SFcnCompatibilityMsg	none
FrameProcessingCompatibilityMsg	error
UniqueDataStoreMsg	none
BusObjectLabelMismatch	warning
RootOutportRequireBusObject	warning
AssertControl	UseLocalSettings
AllowSymbolicDim	on
ModelReferenceVersionMismatchMessage	none
ModelReferenceIOMismatchMessage	none
UnknownTsInhSupMsg	warning
ModelReferenceDataLoggingMessage	warning
ModelReferenceNoExplicitFinalValueMsg	none
ModelReferenceSymbolNameMessage	warning
StateNameClashWarn	none
OperatingPointInterfaceChecksumMismatchMsg	warning
NonCurrentReleaseOperatingPointMsg	error
PregeneratedLibrarySubsystemCodeDiagnostic	warning
SubsystemReferenceDiagnosticForUnitTest	error
InitInArrayFormatMsg	warning
StrictBusMsg	ErrorLevel1
BusNameAdapt	WarnAndRepair
NonBusSignalsTreatedAsBus	none
SFUnusedDataAndEventsDiag	warning
SFUnexpectedBacktrackingDiag	error
SFInvalidInputDataAccessInChartInitDiag	warning
SFNoUnconditionalDefaultTransitionDiag	error

SFTransitionOutsideNaturalParentDiag	warning
SFUnreachableExecutionPathDiag	warning
SFUndirectedBroadcastEventsDiag	warning
SFTransitionActionBeforeConditionDiag	warning
SFOutputUsedAsStateInMooreChartDiag	error
SFTemporalDelaySmallerThanSampleTimeDiag	warning
SFSelfTransitionDiag	warning
SFExecutionAtInitializationDiag	warning
IntegerSaturationMsg	warning
AllowedUnitSystems	all
UnitsInconsistencyMsg	warning
AllowAutomaticUnitConversions	on
RCSCRenamedMsg	warning
RCSCObservableMsg	warning
ForceCombineOutputUpdateInSim	off
UnderSpecifiedDimensionMsg	none
DebugExecutionForFMUViaOutOfProcess	off
ArithmeticOperatorsInVariantConditions	error
VariantConditionMismatch	none
InheritVATfromSVC	warning
VariantConfigNotUsedByTopModel	warning
ParamWriterValidationControl	UseLocalSettings

Table 5.6. warning_system Configuration Set.Components(5)

Property	**Value**
Name	Hardware Implementation
Description	
Components	
ProdBitPerChar	8
ProdBitPerShort	16
ProdBitPerInt	32
ProdBitPerLong	32
ProdBitPerLongLong	64
ProdBitPerFloat	32
ProdBitPerDouble	64
ProdBitPerPointer	32

ProdBitPerSizeT	32
ProdBitPerPtrDiffT	32
ProdLargestAtomicInteger	Long
ProdLargestAtomicFloat	Double
ProdIntDivRoundTo	Zero
ProdEndianess	LittleEndian
ProdWordSize	32
ProdShiftRightIntArith	on
ProdLongLongMode	off
ProdHWDeviceType	ARM Compatible->ARM Cortex
TargetBitPerChar	8
TargetBitPerShort	16
TargetBitPerInt	32
TargetBitPerLong	32
TargetBitPerLongLong	64
TargetBitPerFloat	32
TargetBitPerDouble	64
TargetBitPerPointer	32
TargetBitPerSizeT	32
TargetBitPerPtrDiffT	32
TargetLargestAtomicInteger	Char
TargetLargestAtomicFloat	None
TargetShiftRightIntArith	on
TargetLongLongMode	off
TargetIntDivRoundTo	Undefined
TargetEndianess	Unspecified
TargetWordSize	32
TargetPreprocMaxBitsSint	32
TargetPreprocMaxBitsUint	32
TargetHWDeviceType	Specified
TargetUnknown	off
ProdEqTarget	on
UseEmbeddedCoderFeatures	on
UseSimulinkCoderFeatures	on
HardwareBoardFeatureSet	EmbeddedCoderHSP

Table 5.7. warning_system Configuration Set.Components(6)

Property	Value
Name	Model Referencing
Description	
Components	
UpdateModelReferenceTargets	IfOutOfDateOrStructuralChange
EnableRefExpFcnMdlSchedulingChecks	off
CheckModelReferenceTargetMessage	error
EnableParallelModelReferenceBuilds	off
ParallelModelReferenceErrorOnInvalidPool	on
ParallelModelReferenceMATLABWorkerInit	None
ModelReferenceNumInstancesAllowed	Multi
PropagateVarSize	Infer from blocks in model
ModelDependencies	
ModelReferencePassRootInputsByReference	on
ModelReferenceMinAlgLoopOccurrences	off
PropagateSignalLabelsOutOfModel	on
SupportModelReferenceSimTargetCustomCode	off
UseModelRefSolver	off

Table 5.8. warning_system Configuration Set.Components(7)

Property	Value
Name	Simulation Target
Description	
Components	
SimCustomSourceCode	
SimCustomHeaderCode	
SimCustomInitializer	
SimCustomTerminator	
SimReservedNameArray	
SimUserSources	
SimUserIncludeDirs	
SimUserLibraries	
SimUserDefines	
SimCustomCompilerFlags	

SimCustomLinkerFlags	
SFSimEnableDebug	off
SFSimEcho	on
SimCtrlC	on
SimIntegrity	on
SimUseLocalCustomCode	on
SimParseCustomCode	on
SimAnalyzeCustomCode	off
SimDebugExecutionForCustomCode	off
SimGenImportedTypeDefs	off
CompileTimeRecursionLimit	50
EnableRuntimeRecursion	on
EnableImplicitExpansion	on
MATLABDynamicMemAlloc	on
MATLABDynamicMemAllocThreshold	65536
UsePrecompiledLibraries	Prefer
LegacyBehaviorForPersistentVarInContinuousTime	off
CustomCodeFunctionArrayLayout	
DefaultCustomCodeFunctionArrayLayout	NotSpecified
CustomCodeUndefinedFunction	FilterOut
CustomCodeGlobalsAsFunctionIO	off
DefaultCustomCodeDeterministicFunctions	None
CustomCodeDeterministicFunctions	
SimHardwareAcceleration	generic
SimTargetLang	C
GPUAcceleration	off
SimGPUMallocThreshold	200
SimGPUStackLimitPerThread	1024
SimGPUErrorChecks	off
SimGPUCustomComputeCapability	
SimGPUCompilerFlags	
SimDLTargetLibrary	mkl-dnn
SimDLAutoTuning	on

Table 5.9. warning_system Configuration Set.Components(8)

Property	Value

Name	Code Generation
Description	Embedded Coder
SystemTargetFile	ert.tlc
EmbeddedCoderDictionary	
HardwareBoard	ESP32-WROOM(Arduino Compatible)
ShowCustomHardwareApp	off
ShowEmbeddedHardwareApp	off
TLCOptions	-aInlineSetEventsForThisBaseRateFcn=TLC_FALSE -aSuppressMultiTaskScheduler=TLC_FALSE -aRateBasedStepFcn=0
GenCodeOnly	off
MakeCommand	make_rtw
GenerateMakefile	on
PackageGeneratedCodeAndArtifacts	off
PackageName	
TemplateMakefile	ert_default_tmf
PostCodeGenCommand	codertarget.postCodeGenHookCommand(h)
GenerateReport	off
RTWVerbose	on
RetainRTWFile	off
ProfileTLC	off
TLCDebug	off
TLCCoverage	off
TLCAssert	off
BuiltinFFTWCallback	off
RTWUseLocalCustomCode	on
RTWUseSimCustomCode	off
CustomSourceCode	
CustomHeaderCode	
CustomInclude	
CustomSource	
CustomLibrary	
CustomDefine	
CustomBLASCallback	
CustomLAPACKCallback	
CustomFFTCallback	
CustomInitializer	
CustomTerminator	

Toolchain	ESP32
BuildConfiguration	Faster Builds
CustomToolchainOptions	
IncludeHyperlinkInReport	off
LaunchReport	off
PortableWordSizes	off
CreateSILPILBlock	None
CodeExecutionProfiling	off
CodeExecutionProfileVariable	executionProfile
CodeProfilingSaveOptions	SummaryOnly
CodeProfilingInstrumentation	off
CodeStackProfiling	off
CodeStackProfileVariable	stackProfile
CodeCoverageSettings	warning_system Configuration Set.Components(8).CodeCoverageSettings
SILPILDebugging	off
RemoveFixptWordSizeChecks	off
DataTypeReplacement	CoderTypedefs
CoderTypedefsCompatibility	off
TargetLang	C
GenerateGPUCode	None
HalideCodeGeneration	off
GenerateTraceInfo	off
GenerateTraceReport	off
GenerateTraceReportSl	off
GenerateTraceReportSf	off
GenerateTraceReportEml	off
GenerateWebview	off
GenerateCodeMetricsReport	off
GenerateCodeReplacementReport	off
RTWCompilerOptimization	off
ObjectivePriorities	{Execution efficiency}
RTWCustomCompilerOptimizations	
CheckMdlBeforeBuild	Off
GPUKernelNamePrefix	
GPUDeviceID	-1
GPUMallocMode	discrete

GPUMallocThreshold	200
GPUEnableMemoryManager	off
GPUStackLimitPerThread	1024
GPUcuBLAS	on
GPUcuSOLVER	on
GPUcuFFT	on
GPUErrorChecks	off
GPUComputeCapability	5.0
GPUCustomComputeCapability	
GPUCompilerFlags	
GPUMaximumBlocksPerKernel	0
DLTargetLibrary	none
DLAutoTuning	on
DLDataType	fp32
DLArmComputeVersion	20.02.1
DLArmComputeArch	unspecified
DLLearnablesCompression	None
LargeConstantGeneration	KeepInSourceFiles
LargeConstantThreshold	131072
Components	[warning_system Configuration Set.Components(8).Components(1), warning_system Configuration Set.Components(8).Components(2)]

Table 5.10. warning_system Configuration Set.Components(9)

Property	Value
Description	Simulink Coverage Configuration Component
Components	
Name	Simulink Coverage
CovEnable	off
CovScope	EntireSystem
CovIncludeTopModel	on
RecordCoverage	off
CovPath	/
CovSaveName	covdata
CovCompData	
CovMetricSettings	dwe

CovFilter	
CovHTMLOptions	
CovNameIncrementing	off
CovForceBlockReductionOff	on
CovEnableCumulative	on
CovSaveCumulativeToWorkspaceVar	off
CovSaveSingleToWorkspaceVar	off
CovCumulativeVarName	covCumulativeData
CovCumulativeReport	off
CovSaveOutputData	on
CovOutputDir	slcov_output/$ModelName$
CovDataFileName	$ModelName$_cvdata
CovReportOnPause	on
CovModelRefEnable	off
CovModelRefExcluded	
CovExternalEMLEnable	on
CovSFcnEnable	on
CovBoundaryAbsTol	1.0000e-05
CovBoundaryRelTol	0.0100
CovUseTimeInterval	off
CovStartTime	0
CovStopTime	0
CovMetricStructuralLevel	Decision
CovMetricLookupTable	off
CovMetricSignalRange	off
CovMetricSignalSize	off
CovMetricObjectiveConstraint	off
CovMetricSaturateOnIntegerOverflow	off
CovMetricRelationalBoundary	off
CovLogicBlockShortCircuit	off
CovUnsupportedBlockWarning	on
CovMcdcMode	Masking
CovExcludeInactiveVariants	off

Table 5.11. warning_system Configuration Set.Components(10)

Property	Value

Description	Coder Target
Components	
Name	Coder Target
CoderTargetData	warning_system Configuration Set.Components(10).CoderTargetData
TaskManagerData	
UseSoCProfilerForTargets	off

Table 5.12. warning_system Configuration Set.Components(8).CodeCoverageSettings

Property	Value
TopModelCoverage	off
ReferencedModelCoverage	off
CoverageTool	None

Table 5.13. warning_system Configuration Set.Components(8).Components(1)

Property	Value
Name	Code Appearance
Description	
Components	
ForceParamTrailComments	on
GenerateComments	on
CommentStyle	Auto
IgnoreCustomStorageClasses	off
IgnoreTestpoints	off
MaxIdLength	31
ShowEliminatedStatement	on
OperatorAnnotations	on
SimulinkDataObjDesc	on
SFDataObjDesc	on
MATLABFcnDesc	off
MangleLength	1
SharedChecksumLength	8
CustomSymbolStrGlobalVar	RN$M
CustomSymbolStrType	NR$M_T
CustomSymbolStrField	NM

CustomSymbolStrFcn	RNMF
CustomSymbolStrFcnArg	rt$I$N$M
CustomSymbolStrBlkIO	rtb_NM
CustomSymbolStrTmpVar	NM
CustomSymbolStrMacro	RN$M
CustomSymbolStrUtil	NC
CustomSymbolStrEmxType	emxArray_MN
CustomSymbolStrEmxFcn	emx$M$N
CustomUserTokenString	
CustomCommentsFcn	
DefineNamingRule	None
DefineNamingFcn	
ParamNamingRule	None
ParamNamingFcn	
SignalNamingRule	None
SignalNamingFcn	
InsertBlockDesc	on
AnnotationsInComments	off
InsertPolySpaceComments	off
SimulinkBlockComments	on
BlockCommentType	BlockPathComment
StateflowObjectComments	off
MATLABSourceComments	off
EnableCustomComments	off
InternalIdentifier	Shortened
InlinedPrmAccess	Literals
ReqsInCode	off
UseSimReservedNames	off
ReservedNameArray	
EnumMemberNameClash	error

Table 5.14. warning_system Configuration Set.Components(8).Components(2)

Property	**Value**
Name	Target
Description	
Components	

IsERTTarget	on
TargetLibSuffix	
TargetPreCompLibLocation	
TargetLangStandard	C89/C90 (ANSI)
CodeReplacementLibrary	None
UtilityFuncGeneration	Auto
MultiwordTypeDef	System defined
MultiwordLength	2048
DynamicStringBufferSize	2048
GenerateFullHeader	on
InferredTypesCompatibility	off
ExistingSharedCode	
GenerateSampleERTMain	off
GenerateTestInterfaces	off
ModelReferenceCompliant	on
ParMdlRefBuildCompliant	on
CompOptLevelCompliant	on
ConcurrentExecutionCompliant	on
IncludeMdlTerminateFcn	on
CombineOutputUpdateFcns	on
CombineSignalStateStructs	off
GroupInternalDataByFunction	off
SuppressErrorStatus	off
IncludeFileDelimiter	Auto
ERTCustomFileBanners	on
SupportAbsoluteTime	on
LogVarNameModifier	rt_
MatFileLogging	off
MultiInstanceERTCode	off
CodeInterfacePackaging	Nonreusable function
PurelyIntegerCode	off
SupportNonFinite	on
SupportComplex	on
SupportContinuousTime	on
SupportNonInlinedSFcns	on
RemoveDisableFunc	off

RemoveResetFunc	on
SupportVariableSizeSignals	on
ParenthesesLevel	Nominal
CastingMode	Nominal
ModelStepFunctionPrototypeControlCompliant	on
CPPClassGenCompliant	on
GRTInterface	off
GenerateAllocFcn	off
UseToolchainInfoCompliant	on
GenerateSharedConstants	on
LUTObjectStructOrderExplicitValues	Size,Breakpoints,Table
LUTObjectStructOrderEvenSpacing	Size,Breakpoints,Table
ArrayLayout	Column-major
UnsupportedSFcnMsg	error
ERTHeaderFileRootName	RE
ERTSourceFileRootName	RE
ERTDataFileRootName	$R_data
InstructionSetExtensions	{None}
InstructionSetFMA	off
OptimizeReductions	off
IsSLRTTarget	off
HeaderGuardPrefix	
LogToMDFFile	off
DSAsUniqueAccess	off
ExtMode	off
ExtModeTransport	3
ExtModeStaticAlloc	on
ExtModeAutomaticAllocSize	off
ExtModeMaxTrigDuration	10
ExtModeStaticAllocSize	65536
ExtModeTesting	off
ExtModeMexFile	ext_xcp
ExtModeMexArgs	
ExtModeIntrfLevel	Level2 - Open
TargetOS	BareBoardExample
MultiInstanceErrorCode	Error

RootIOFormat	Individual arguments
RTWCAPISignals	off
RTWCAPIParams	off
RTWCAPIStates	on
RTWCAPIRootIO	on
ERTSrcFileBannerTemplate	ert_code_template.cgt
ERTHdrFileBannerTemplate	ert_code_template.cgt
ERTDataSrcFileTemplate	ert_code_template.cgt
ERTDataHdrFileTemplate	ert_code_template.cgt
ERTCustomFileTemplate	codertarget_file_process.tlc
EnableDataOwnership	off
SignalDisplayLevel	10
ParamTuneLevel	10
GlobalDataDefinition	Auto
DataDefinitionFile	global.c
GlobalDataReference	Auto
ERTFilePackagingFormat	Modular
RateTransitionBlockCode	Inline
DataReferenceFile	global.h
PreserveExpressionOrder	off
PreserveIfCondition	off
ConvertIfToSwitch	on
PreserveExternInFcnDecls	on
PreserveStaticInFcnDecls	on
SuppressUnreachableDefaultCases	on
EnableSignedLeftShifts	on
EnableSignedRightShifts	on
ImplementImageWithCVMat	off
IndentStyle	K&R
IndentSize	2
NewlineStyle	Default
MaxLineWidth	80
EnableUserReplacementTypes	off
ReplacementTypes	warning_system Configuration Set.Components(8).Components(2).ReplacementTypes

MaxIdInt64	MAX_int64_T
MinIdInt64	MIN_int64_T
MaxIdUint64	MAX_uint64_T
MaxIdInt32	MAX_int32_T
MinIdInt32	MIN_int32_T
MaxIdUint32	MAX_uint32_T
MaxIdInt16	MAX_int16_T
MinIdInt16	MIN_int16_T
MaxIdUint16	MAX_uint16_T
MaxIdInt8	MAX_int8_T
MinIdInt8	MIN_int8_T
MaxIdUint8	MAX_uint8_T
BooleanTrueId	true
BooleanFalseId	false
TypeLimitIdReplacementHeaderFile	
ArrayContainerType	C-style array
DynamicArrayContainerType	std::vector

Table 5.15. warning_system Configuration Set.Components(10).CoderTargetData

Field	Value
UseCoderTarget	true
TargetHardware	ESP32-WROOM(Arduino Compatible)
ExtModeProtocolInfo	warning_system Configuration Set.Components(10).CoderTargetData.ExtModeProtocolInfo
ConnectionInfo	warning_system Configuration Set.Components(10).CoderTargetData.ConnectionInfo
ExtMode	warning_system Configuration Set.Components(10).CoderTargetData.ExtMode
RTOS	FreeRTOS
RTOSBaseRateTaskPriority	14
DetectTaskOverruns	0
Scheduler_interrupt_source	0
ESP32	warning_system Configuration Set.Components(10).CoderTargetData.ESP32
Runtime	warning_system Configuration Set.Components(10).CoderTargetData.Runtime

HostBoardConnection	warning_system Configuration Set.Components(10).CoderTargetData.HostBoardConnection
ConnectedIO	warning_system Configuration Set.Components(10).CoderTargetData.ConnectedIO
OverRunDetection	warning_system Configuration Set.Components(10).CoderTargetData.OverRunDetection
AnalogInRefVoltage	warning_system Configuration Set.Components(10).CoderTargetData.AnalogInRefVoltage
Serial	warning_system Configuration Set.Components(10).CoderTargetData.Serial
I2C	warning_system Configuration Set.Components(10).CoderTargetData.I2C
SPI	warning_system Configuration Set.Components(10).CoderTargetData.SPI
Wifi	warning_system Configuration Set.Components(10).CoderTargetData.Wifi
ThingSpeak	warning_system Configuration Set.Components(10).CoderTargetData.ThingSpeak
PWM	warning_system Configuration Set.Components(10).CoderTargetData.PWM
BLE	warning_system Configuration Set.Components(10).CoderTargetData.BLE
MQTT	warning_system Configuration Set.Components(10).CoderTargetData.MQTT
Http	warning_system Configuration Set.Components(10).CoderTargetData.Http
IOBlocksMode	deployed
DataVersion	2016.02
DashboardCodegenInfo	warning_system Configuration Set.Components(10).CoderTargetData.DashboardCodegenInfo

Table 5.16. warning_system Configuration Set.Components(8).Components(2).ReplacementTypes

Field	Value
double	
single	
int32	
int16	
int8	
uint32	
uint16	
uint8	
boolean	
int	
uint	
char	

uint64	
int64	

Table 5.17. warning_system Configuration Set.Components(10).CoderTargetData.ExtModeProtocolInfo

Field	Value
XCPonSerial	warning_system Configuration Set.Components(10).CoderTargetData.ExtModeProtocolInfo.XCPonSerial
XCPonWiFi	warning_system Configuration Set.Components(10).CoderTargetData.ExtModeProtocolInfo.XCPonWiFi

Table 5.18. warning_system Configuration Set.Components(10).CoderTargetData.ConnectionInfo

Field	Value
XCPonSerial	warning_system Configuration Set.Components(10).CoderTargetData.ConnectionInfo.XCPonSerial
XCPonWiFi	warning_system Configuration Set.Components(10).CoderTargetData.ConnectionInfo.XCPonWiFi

Table 5.19. warning_system Configuration Set.Components(10).CoderTargetData.ExtMode

Field	Value
Configuration	XCP on Serial
COMPortBaud	115200
targetPollingTimeMode	Automatically
targetPollingTime	2
UseRealtimeStampLogging	0

Table 5.20. warning_system Configuration Set.Components(10).CoderTargetData.ESP32

Field	Value
ESP32_board	ESP32-WROOM-DevKitV1(30 pin)
BootButtonPressCheckbox	0

Table 5.21. warning_system Configuration Set.Components(10).CoderTargetData.Runtime

Field	Value
BuildAction	Build
DisableParallelBuild	0
ForceBuildStaticLibrary	0
VerboseOutput	true

Table 5.22. warning_system Configuration Set.Components(10).CoderTargetData.HostBoardConnection

Field	Value
AppDownload_port_source	Manually Specify
AppDownload_COMPort_select	
AppDownload_COMPort_specify	6
AppDownload_baud	921600(Default)
AppDownload_baud_specify	9600
ConnectedIO_serial_port	Serial 0
ConnectedIO_port_source1	Manually Specify
ConnectedIO_port_source2	Manually Select
ConnectedIO_COMPort_select	
ConnectedIO_COMPort_specify	6
ConnectedIO_baud	115200
ConnectedIO_baud_specify	
ExternalMode_serial_port	Serial 0
ExternalMode_port_source1	Manually Specify
ExternalMode_port_source2	Manually Select
ExternalMode_COMPort_select	
ExternalMode_COMPort_specify	6
ExternalMode_baud	115200
ExternalMode_baud_specify	

Table 5.23. warning_system Configuration Set.Components(10).CoderTargetData.ConnectedIO

Field	Value
ConnectedIOMode	Auto

ActionOnOverrun	warning

Table 5.24. warning_system Configuration Set.Components(10).CoderTargetData.OverRunDetection

Field	Value
Enable_overrun_detection	0
Digital_output_to_set_on_overrun	13

Table 5.25. warning_system Configuration Set.Components(10).CoderTargetData.AnalogInRefVoltage

Field	Value
Analog_input_reference_voltage	0

Table 5.26. warning_system Configuration Set.Components(10).CoderTargetData.Serial

Field	Value
Serial0_baud_rate	921600
Serial0_baud_specify	921600
Serial0_Config	SERIAL_8N1
Serial0_tx_pin	1
Serial0_rx_pin	3
Serial1_baud_rate	9600
Serial1_baud_specify	921600
Serial1_Config	SERIAL_8N1
Serial1_tx_pin	33
Serial1_rx_pin	32
Serial2_baud_rate	9600
Serial2_baud_specify	921600
Serial2_Config	SERIAL_8N1
Serial2_tx_pin	17
Serial2_rx_pin	16

Table 5.27. warning_system Configuration Set.Components(10).CoderTargetData.I2C

Field	Value
I2C0BusSpeedHz	400000
I2C0_SDA_Pin	21
I2C0_SCL_Pin	22

Table 5.28. warning_system Configuration Set.Components(10).CoderTargetData.SPI

Field	Value
SPI0_clock_out_frequency	4000
SPI0_mode	Mode 0 - Clock Polarity 0, Clock Phase 0
SPI0_bitorder	MSB first
SPI0_SDO_Pin	23
SPI0_SDI_Pin	19
SPI0_SCK_Pin	18
SDSlaveSelect	5
SPI1_clock_out_frequency	4000
SPI1_mode	Mode 0 - Clock Polarity 0, Clock Phase 0
SPI1_bitorder	MSB first
SPI1_SDO_Pin	13
SPI1_SDI_Pin	12
SPI1_SCK_Pin	14

Table 5.29. warning_system Configuration Set.Components(10).CoderTargetData.Wifi

Field	Value
Wifi_SSID	yourNetwork
Set_Wifi_Encryption	None
Wifi_WEP_key	D0D0DEADF00DABBADEAFBEADED
Wifi_WEP_key_index	0
Wifi_WPA_password	secretPassword
Wifi_WPA2_username	username
Wifi_WPA2_password	password
IP_Address_Assignment	DHCP

Wifi_ip_address	192.168.1.20
Wifi_dns_server_address	192.168.1.0
Wifi_gateway_address	192.168.1.0
Wifi_subnet_mask	255.255.255.0

Table 5.30. warning_system Configuration Set.Components(10).CoderTargetData.ThingSpeak

Field	Value
Enable_CustomServer	0
IP_address	184.106.153.149
Port	80

Table 5.31. warning_system Configuration Set.Components(10).CoderTargetData.PWM

Field	Value
PWM_Default_Frequency	1000

Table 5.32. warning_system Configuration Set.Components(10).CoderTargetData.BLE

Field	Value
Peripheral_device_name	Arduino
Peripheral_local_name	Arduino
Advertising_interval	160
Min_connection_interval	6
Max_connection_interval	3200

Table 5.33. warning_system Configuration Set.Components(10).CoderTargetData.MQTT

Field	Value
BrokerService	ThingSpeak
BrokerAddress	mqtt3.thingspeak.com
BrokerPort	1883
Username	
Password	

ClientID	

Table 5.34. warning_system Configuration Set.Components(10).CoderTargetData.Http

Field	Value
Enable_custom_ca_certificate	0
Custom_ca_certificate_file	

Table 5.35. warning_system Configuration Set.Components(10).CoderTargetData.DashboardCodegenInfo

Field	Value
enableCodegen	false
circulargauge	warning_system Configuration Set.Components(10).CoderTargetData.DashboardCodegenInfo.circulargauge
displayblock	warning_system Configuration Set.Components(10).CoderTargetData.DashboardCodegenInfo.displayblock
pushbutton	warning_system Configuration Set.Components(10).CoderTargetData.DashboardCodegenInfo.pushbutton

Table 5.36. warning_system Configuration Set.Components(10).CoderTargetData.ExtModeProtocolInfo.XCPonSerial

Field	Value
HostInterface	Simulink
LoggingBufferAuto	true
LoggingBufferSize	250
LoggingBufferNum	3
MaxContigSamples	10

Table 5.37. warning_system Configuration Set.Components(10).CoderTargetData.ExtModeProtocolInfo.XCPonWiFi

Field	Value
HostInterface	Simulink
LoggingBufferAuto	true
LoggingBufferSize	250
LoggingBufferNum	3

MaxContigSamples	10

Table 5.38. warning_system Configuration Set.Components(10).CoderTargetData.ConnectionInfo.XCPonSerial

Field	Value
Baudrate	codertarget.arduinobase.registry.getBaudRate
COMPort	codertarget.arduinobase.internal.getExternalModeMexArgs('Serial')
Verbose	false

Table 5.39. warning_system Configuration Set.Components(10).CoderTargetData.ConnectionInfo.XCPonWiFi

Field	Value
IPAddress	codertarget.arduinobase.internal.getExternalModeMexArgs('Wifi')
Port	17725
Verbose	false

Table 5.40. warning_system Configuration Set.Components(10).CoderTargetData.DashboardCodegenInfo.circulargauge

Field	Value
Codegen	true
Blockclass	codertarget.targetHiddenBlkInsert.internal.circularGauge
RegFcn	codertarget.arduinobase.blocks.registerDashboardBlk
ValidateFcn	codertarget.targetHiddenBlkInsert.internal.isDashboardBlockCodegenEnabled

Table 5.41. warning_system Configuration Set.Components(10).CoderTargetData.DashboardCodegenInfo.displayblock

Field	Value
Codegen	true
Blockclass	codertarget.targetHiddenBlkInsert.internal.lcdTextDisplay
RegFcn	codertarget.arduinobase.blocks.registerDashboardBlk
ValidateFcn	codertarget.targetHiddenBlkInsert.internal.isDashboardBlockCodegenEnabled

Table 5.42. warning_system Configuration Set.Components(10).CoderTargetData.DashboardCodegenInfo.pushbutton

Field	Value
Codegen	true
Blockclass	codertarget.targetHiddenBlkInsert.internal.pushButton
RegFcn	codertarget.arduinobase.blocks.registerDashboardBlk
ValidateFcn	codertarget.targetHiddenBlkInsert.internal.isDashboardBlockCodegenEnabled

Chapter 6. Glossary

Atomic Subsystem. A subsystem treated as a unit by an implementation of the design documented in this report. The implementation computes the outputs of all the blocks in the atomic subsystem before computing the next block in the parent system's block execution order (sorted list).

Block Diagram. A Simulink block diagram represents a set of simultaneous equations that relate a system or subsystem's inputs to its outputs as a function of time. Each block in the diagram represents an equation of the form y = f(t, x, u) where t is the current time, u is a block input, y is a block output, and x is a system state (see the Simulink documentation for information on the functions represented by the various types of blocks that make up the diagram). Lines connecting the blocks represent dependencies among the blocks, i.e., inputs whose current values are the outputs of other blocks. An implementation of a design described in this document computes a root or atomic system's outputs at each time step by computing the outputs of the blocks in an order determined by block input/output dependencies.

Block Parameter. A variable that determines the output of a block along with its inputs, for example, the gain parameter of a Gain block.

Block Execution Order. The order in which Simulink evaluates blocks during simulation of a model. The block execution order determined by Simulink ensures that a block executes only after all blocks on whose outputs it depends are executed.

Checksum. A number that indicates whether different versions of a model or atomic subsystem differ functionally or only cosmetically. Different checksums for different versions of the same model or subsystem indicate that the versions differ functionally.

Design Variable. A symbolic (MATLAB) variable or expression used as the value of a block parameter. Design variables allow the behavior of the model to be altered by altering the value of the design variable.

Enumeration Type. Enumerated data is data that is restricted to a finite set of values. An enumerated data type is a MATLAB® class that defines a set of enumerated values. Each enumerated value consists of an enumerated name and an underlying integer which the software uses internally and in generated code.

Signal. A block output, so-called because block outputs typically vary with time.

Virtual Subsystem. A subsystem that is purely graphical, i.e., is intended to reduce the visual complexity of the block diagram of which it is a subsystem. An implementation of the design treats the blocks in the subsystem as part of the first nonvirtual ancestor of the virtual subsystem (see Atomic Subsystem).

Chapter 7. About this Report

Report Overview

This report describes the design of the warning_system system. The report was generated automatically from a Simulink model used to validate the design. It contains the following sections:

Model Version. Specifies information about the version of the model from which this design description was generated. Includes the model checksum, a number that indicates whether different versions of the model differ functionally or only cosmetically. Different checksums for different versions indicate that the versions differ functionally.

Root System. Describes the design's root system.

Subsystems. Describes each of the design's subsystems.

Design Variables. Describes system design variables, i.e., MATLAB variables and expressions used as block parameter values.

Enumeration Type. Describes the enumeration types used by this model.

System Model Configuration. Lists the configuration parameters, e.g., start and stop time, of the model used to simulate the system described by this report.

Glossary. Defines Simulink terms used in this report.

Root System Description

This section describes a design's root system. It contains the following sections:

Diagram. Simulink block diagram that represents the algorithm used to compute the root system's outputs.

Description. Description of the root system. This section appears only if the model's root system has a Documentation property or a Doc block.

Interface. Name, data type, width, and other properties of the root system's input and output signals. The number of the block port that outputs the signal appears in angle brackets appended to the signal name. This section appears only if the root system has input or output ports.

Blocks. This section has two subsections:

- **Parameters.** Describes key parameters of blocks in the root system. This section also includes graphical and/or tabular representations of lookup table data used by lookup table blocks, i.e., blocks that use lookup tables to compute their outputs.

- **Block Execution Order.** Order in which blocks must be executed at each time step in order to ensure that each block's inputs are available when it executes.

State Charts. Describes state charts used in the root system. This section appears only if the root system contains Stateflow blocks.

Subsystem Descriptions

This section describes a design's subsystems. Each subsystem description contains the following sections:
Checksum. This section appears only if the subsystem is an atomic subsystem. The checksum indicates whether the version of the model subsystem used to generate this report differs functionally from other versions of the model subsystem. If two model checksums differ, the corresponding versions of the model differ functionally.
Diagram. Simulink block diagram that graphically represents the algorithm used to compute the subsystem's outputs.
Description. Description of the subsystem. This section appears only if the subsystem has a Documentation property or contains a Doc block.
Interface. Name, data type, width, and other properties of the subsystem's input and output signals. The number of the block port that outputs the signal appears in angle brackets appended to the signal name. This section appears only if the subsystem is atomic and has input or output ports.
Blocks. Blocks that this subsystem contains. This section has two subsections:

- **Parameters.** Key parameters of blocks in the subsystem. This section also includes graphical and/or tabular representations of lookup table data used by lookup table blocks, blocks that use lookup tables to compute their outputs.
- **Block Execution Order.** Order in which the subsystem's blocks must be executed at each time step in order to ensure that each block's inputs are available when the block executes .This section appears only if the subsystem is atomic. Note: in Acrobat(PDF) reports, the number in square brackets next to the block name is a hyperlink to the block parameter table. The number has no model significance.

State Charts. Describes state charts used in the subsystem. This section appears only if the root system contains Stateflow blocks.

State Chart Descriptions

This section describes the state machines used by Stateflow blocks to compute their outputs, i.e., Stateflow blocks. Each state machine description contains the following sections:
Chart. Diagram representing the state machine.
States. Describes the state machine's states. Each state description includes the state's diagram and diagrams and/or descriptions of graphical functions, Simulink functions, truth tables, and MATLAB functions parented by the state.
Transitions. Transitions between the state machine's states. Each transition description specifies the values of key transition properties. Appears only if a transition has properties that do not appear on the chart.
Junctions. Transition junctions. Each junction description specifies the values of key junction properties. Appears only if a junction has properties that do not appear on the chart.

Events. Events that trigger state transitions. Each event description specifies the values of key event properties.
Data. Data types and other properties of the Stateflow block's inputs, outputs, and other state machine data.
Targets. Executable implementations of the state machine used to compute the outputs of the corresponding Stateflow block.
MATLAB Supporting Functions. List of functions invoked by MATLAB functions defined in the chart.